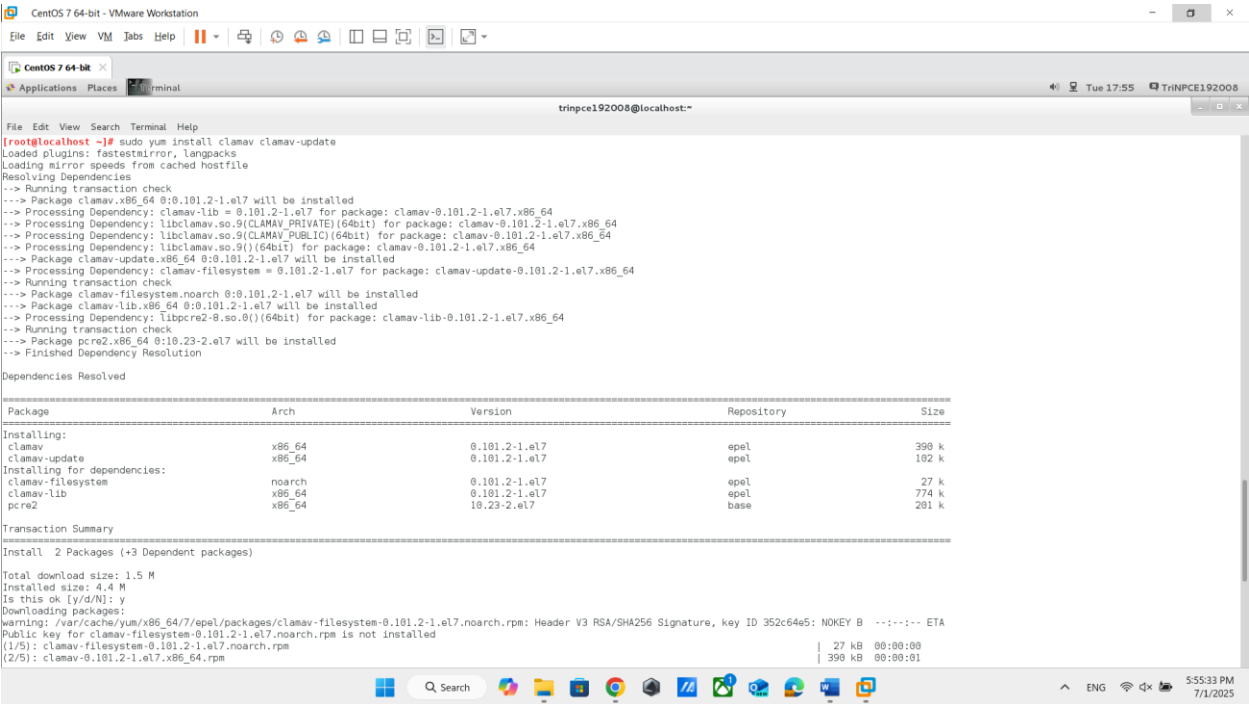
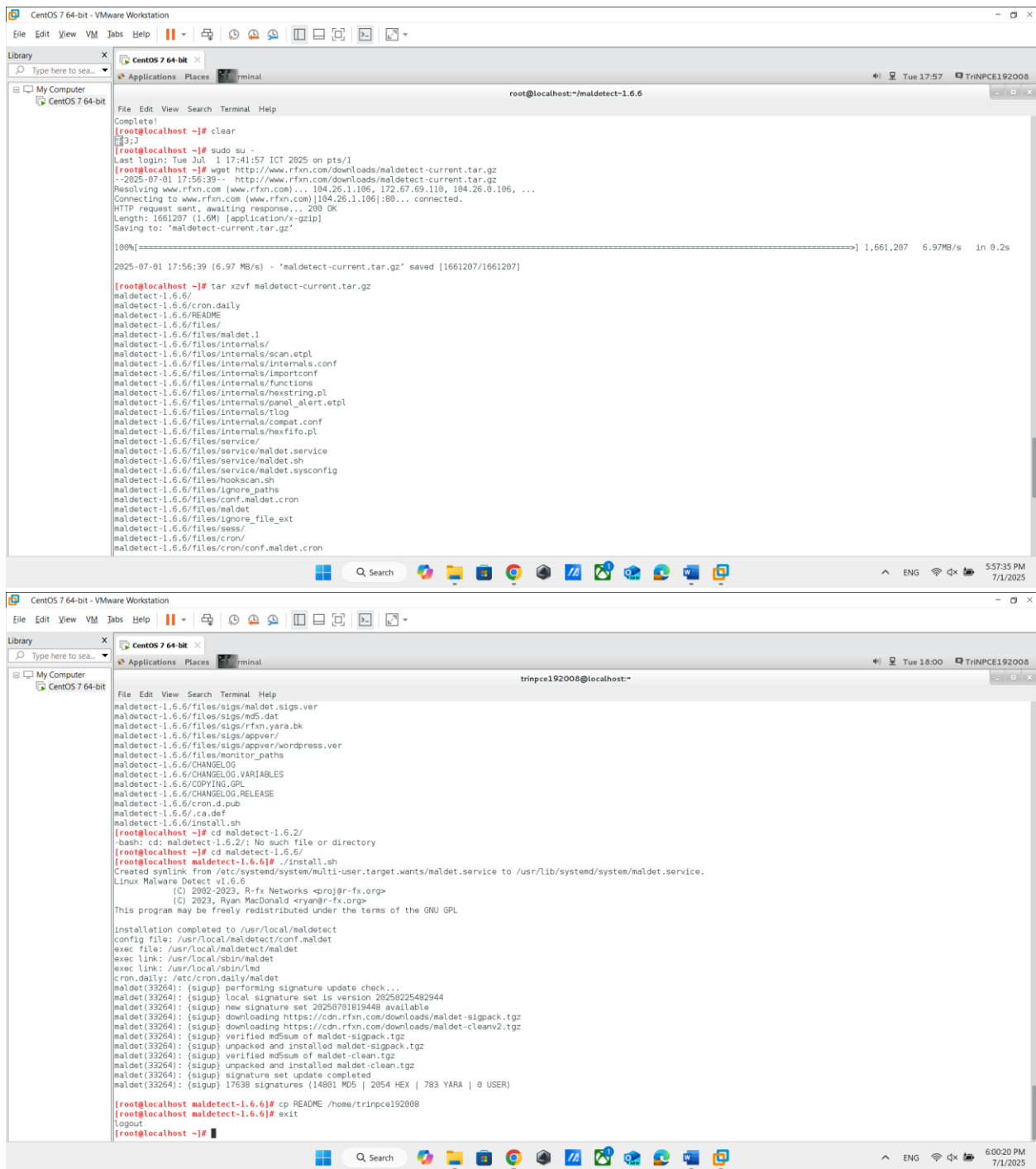


Nguyễn Phạm Trí CE192008 LAB 7

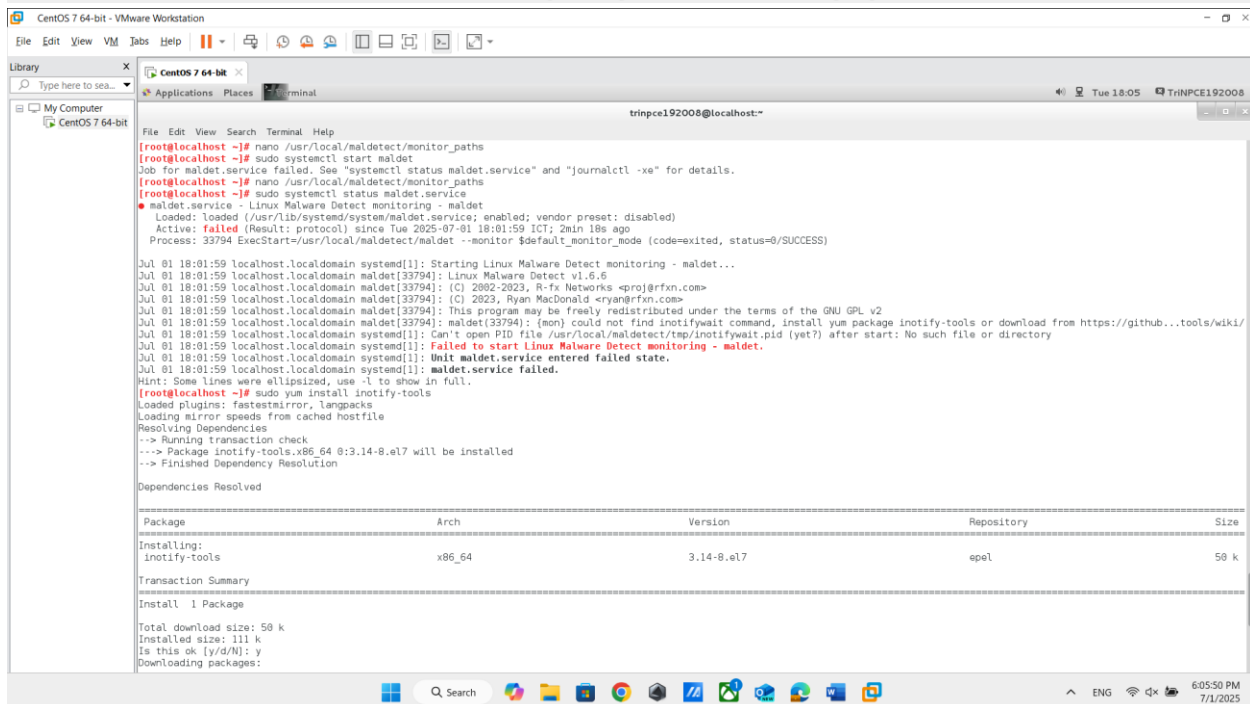
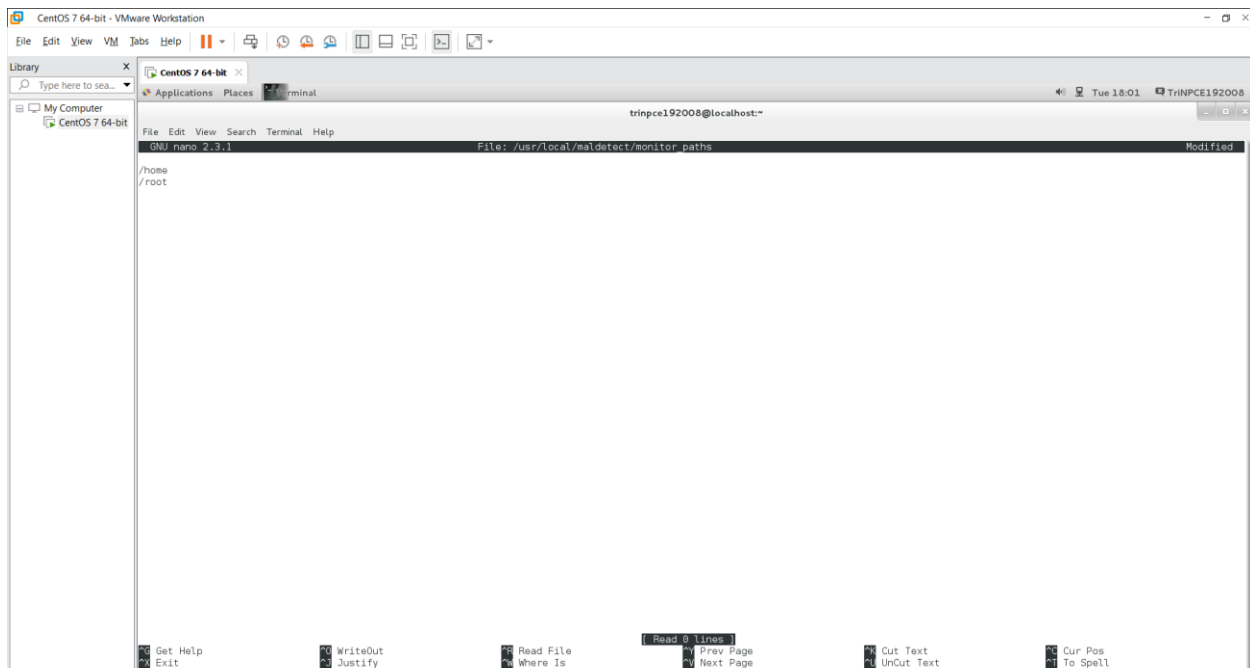
Installing ClamAV and maldet



ảnh trên: cài đặt clamAV



2 ảnh trên: tải và cài đặt maldet



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost:~
File Edit View Search Terminal Help
inotify-tools x86_64 3.14-8.el7 epel 58 k
Transaction Summary
Install 1 Package
Total download size: 58 k
Installed size: 111 k
Is this ok [y/d/N]: y
Downloading packages:
inotify-tools-3.14-8.el7.x86_64.rpm | 58 kB 00:00:01
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : inotify-tools-3.14-8.el7.x86_64 1/1
Verifying : inotify-tools-3.14-8.el7.x86_64 1/1
Installed:
inotify-tools.x86_64 0:3.14-8.el7
Complete!
[root@localhost ~]# sudo systemctl restart maldet
[root@localhost ~]# sudo systemctl status maldet
● maldet.service - Linux Malware Detect monitoring - maldet
   Loaded: loaded (/usr/lib/systemd/system/maldet.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2025-07-01 18:05:30 ICT; 6s ago
     Process: 34137 ExecStart=/usr/local/maldetect/maldet --monitor $default_monitor_mode (code=exited, status=0/SUCCESS)
    Main PID: 34745 (inotifywait)
      CGroup: /system.slice/maldet.service
              └─34745 /usr/bin/inotifywait -r --fromfile /usr/local/maldetect/scan/inotify.paths.34137 --exclude (/var/tmp/mysql.sock$)~/tmp/mysql.sock$~/var/cache/bugagent/md9.cache.data$...
                └─34758 bash /usr/local/maldetect/maldet --monitor users
                  └─34769 sleep 15

Jul 01 18:05:28 localhost.localdomain maldet[34137]: (C) 2023, Ryan MacDonald <ryan@rxfn.com>
Jul 01 18:05:28 localhost.localdomain maldet[34137]: This program may be freely redistributed under the terms of the GNU GPL v2
Jul 01 18:05:28 localhost.localdomain maldet[34137]: maldet[34137]: (mon) set inotify max user watches to 49152
Jul 01 18:05:28 localhost.localdomain maldet[34137]: maldet[34137]: (mon) added /dev/shm to inotify monitoring array
Jul 01 18:05:28 localhost.localdomain maldet[34137]: maldet[34137]: (mon) added /var/tmp to inotify monitoring array
Jul 01 18:05:28 localhost.localdomain maldet[34137]: maldet[34137]: (mon) added /tmp to inotify monitoring array
Jul 01 18:05:28 localhost.localdomain maldet[34137]: maldet[34137]: (mon) inotify startup successful (pid: 34745)
Jul 01 18:05:30 localhost.localdomain maldet[34137]: maldet[34137]: (mon) inotify monitoring log: /usr/local/maldetect/logs/inotify_log
Jul 01 18:05:30 localhost.localdomain systemd[1]: Started Linux Malware Detect monitoring - maldet.
[root@localhost ~]#
```

3 ảnh trên: cài đặt thêm công cụ inotifywait để kích hoạt maldetect

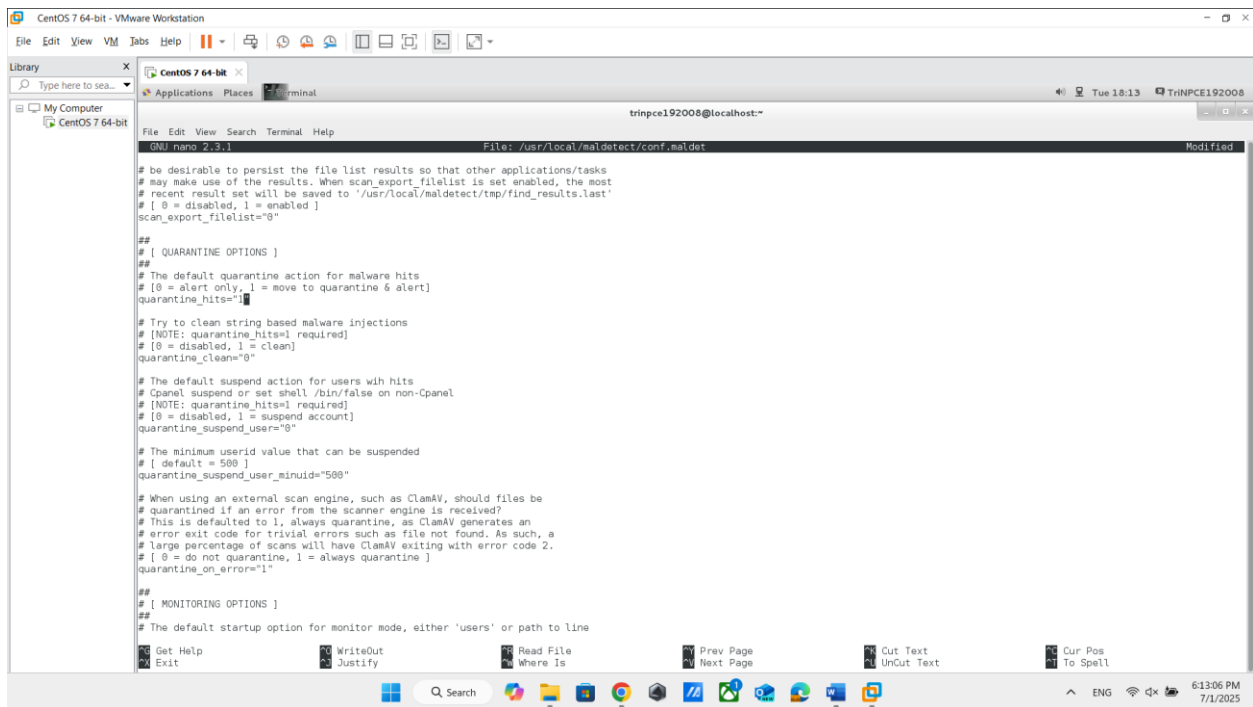
```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost:~
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /usr/local/maldetect/conf/maldet Modified
##
# Linux Malware Detect v1.6.6
# (C) 2002-2025, R-fx Networks <proj@r-fx.org>
# (C) 2025, Ryan MacDonald <ryan@r-fx.org>
# This program may be freely redistributed under the terms of the GNU GPL v2
##
##
# [ General Options ]
##
# Enable or disable e-mail alerts, this includes application version
# alerts as well as automated/manual scan reports. On-demand reports
# can still be sent using '--report SCANID user@domain.com'.
# [ 0 = disabled, 1 = enabled ]
email_alert="1"

# The destination e-mail addresses for automated/manual scan reports
# and application version alerts.
# [ multiple addresses comma (,) spaced ]
email_addr="trinpce192008@gmail.com"

# Ignore e-mail alerts for scan reports in which all malware hits
# have been automatically and successfully cleaned.
# [ 0 = disabled, 1 = enabled ]
email_ignore_clean="1"

# Enable user alerts for specific web hosting control panels. If hits are detected,
# attempt to determine the web hosting control in use, if any. If a control
# panel is detected, determine the user contact information from the panel's
# toolset and send an email summary of the detected hits to that user.
# The list of hits will be limited to files owned by the panel user/account in question.
# Disabling alerts globally with email_alert will also disable this function.
email_panel_user_alerts="0"

# The from header that will be set on alerts to control panel users. This should
# be set by any web hosts that will be supporting the control panel users/accounts
# on this server.
email_panel_from="you@example.com"
```



```
File Edit View Search Terminal Help
File: /usr/local/maldetect/conf.maldet Modified
GNU nano 2.3.1
# be desirable to persist the file list results so that other applications/tasks
# may make use of the results. When scan_export_filelist is set enabled, the most
# recent result set will be saved to '/usr/local/maldetect/tmp/find_results.last'
# [ 0 = disabled, 1 = enabled ]
scan_export_filelist="0"

##
## [ QUARANTINE OPTIONS ]
##
# The default quarantine action for malware hits
# [0 = alert only, 1 = move to quarantine & alert]
quarantine_hits="1"

# Try to clean string based malware injections
# [NOTE: quarantine_hits=1 required]
# [0 = disabled, 1 = clean]
quarantine_clean="0"

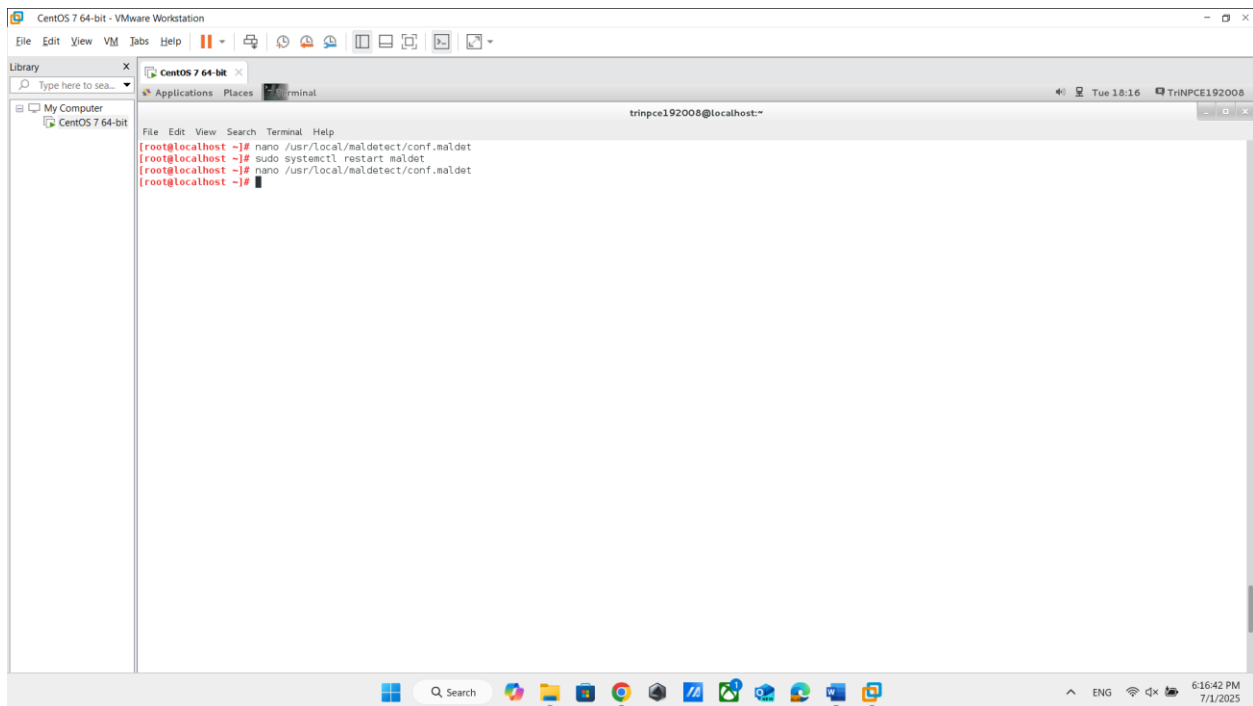
# The default suspend action for users with hits
# Cpanel suspend or set shell /bin/false on non-Cpanel
# [NOTE: quarantine_hits=1 required]
# [0 = disabled, 1 = suspend account]
quarantine_suspend_user="0"

# The minimum userid value that can be suspended
# [ default = 500 ]
quarantine_suspend_user_minuid="500"

# When using an external scan engine, such as ClamAV, should files be
# quarantined if an error from the scanner engine is received?
# This is defaulted to 1, always quarantine, as ClamAV generates an
# error exit code for trivial errors such as file not found. As such, a
# large percentage of scans will have ClamAV exiting with error code 2.
# [ 0 = do not quarantine, 1 = always quarantine ]
quarantine_on_error="1"

##
## [ MONITORING OPTIONS ]
##
# The default startup option for monitor mode, either 'users' or path to line

Get Help WriteOut Read File Prev Page Cut Text
Exit Justify Where Is Next Page UnCut Text
Cur Pos To Spell
```



```
File Edit View Search Terminal Help
File: /usr/local/maldetect/conf.maldet Modified
GNU nano 2.3.1
# be desirable to persist the file list results so that other applications/tasks
# may make use of the results. When scan_export_filelist is set enabled, the most
# recent result set will be saved to '/usr/local/maldetect/tmp/find_results.last'
# [ 0 = disabled, 1 = enabled ]
scan_export_filelist="0"

##
## [ QUARANTINE OPTIONS ]
##
# The default quarantine action for malware hits
# [0 = alert only, 1 = move to quarantine & alert]
quarantine_hits="1"

# Try to clean string based malware injections
# [NOTE: quarantine_hits=1 required]
# [0 = disabled, 1 = clean]
quarantine_clean="0"

# The default suspend action for users with hits
# Cpanel suspend or set shell /bin/false on non-Cpanel
# [NOTE: quarantine_hits=1 required]
# [0 = disabled, 1 = suspend account]
quarantine_suspend_user="0"

# The minimum userid value that can be suspended
# [ default = 500 ]
quarantine_suspend_user_minuid="500"

# When using an external scan engine, such as ClamAV, should files be
# quarantined if an error from the scanner engine is received?
# This is defaulted to 1, always quarantine, as ClamAV generates an
# error exit code for trivial errors such as file not found. As such, a
# large percentage of scans will have ClamAV exiting with error code 2.
# [ 0 = do not quarantine, 1 = always quarantine ]
quarantine_on_error="1"

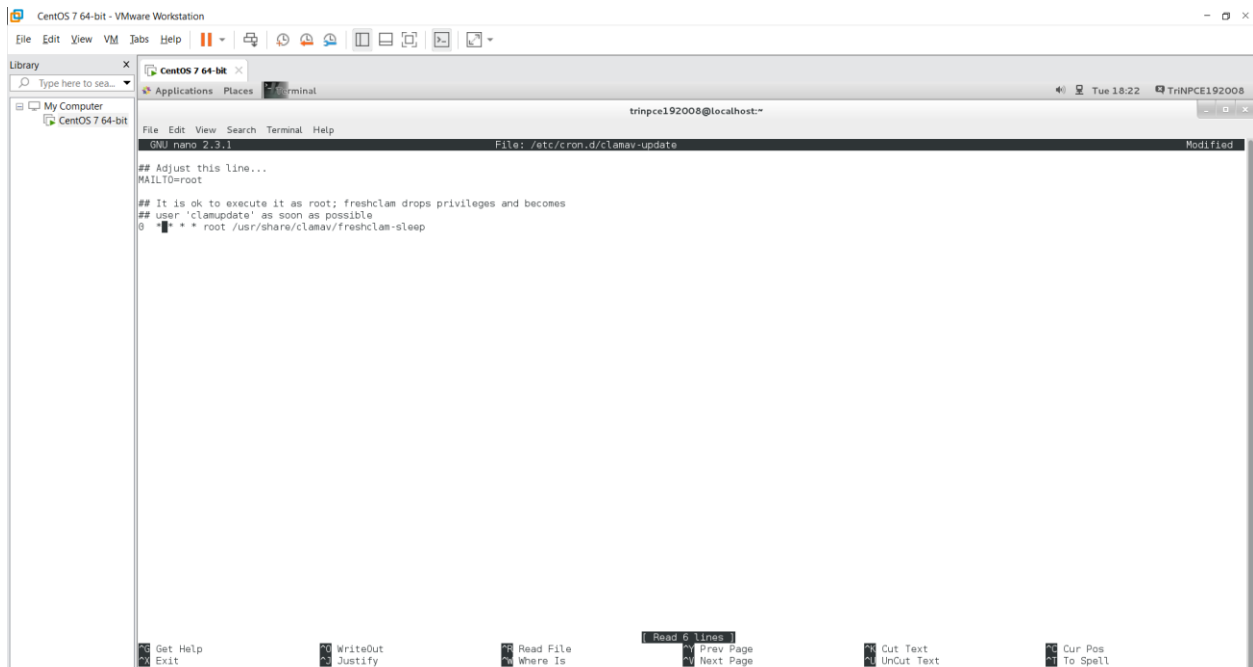
##
## [ MONITORING OPTIONS ]
##
# The default startup option for monitor mode, either 'users' or path to line

Get Help WriteOut Read File Prev Page Cut Text
Exit Justify Where Is Next Page UnCut Text
Cur Pos To Spell
```

3 ảnh trên: sửa file /usr/local/maldetect/conf.maldet: email_alert="1", email_addr
cấu hình gửi email và đổi thuộc tính quarantine_hits="1" để bật tính năng cách ly
(quarantine)

Updating ClamAV and maldet

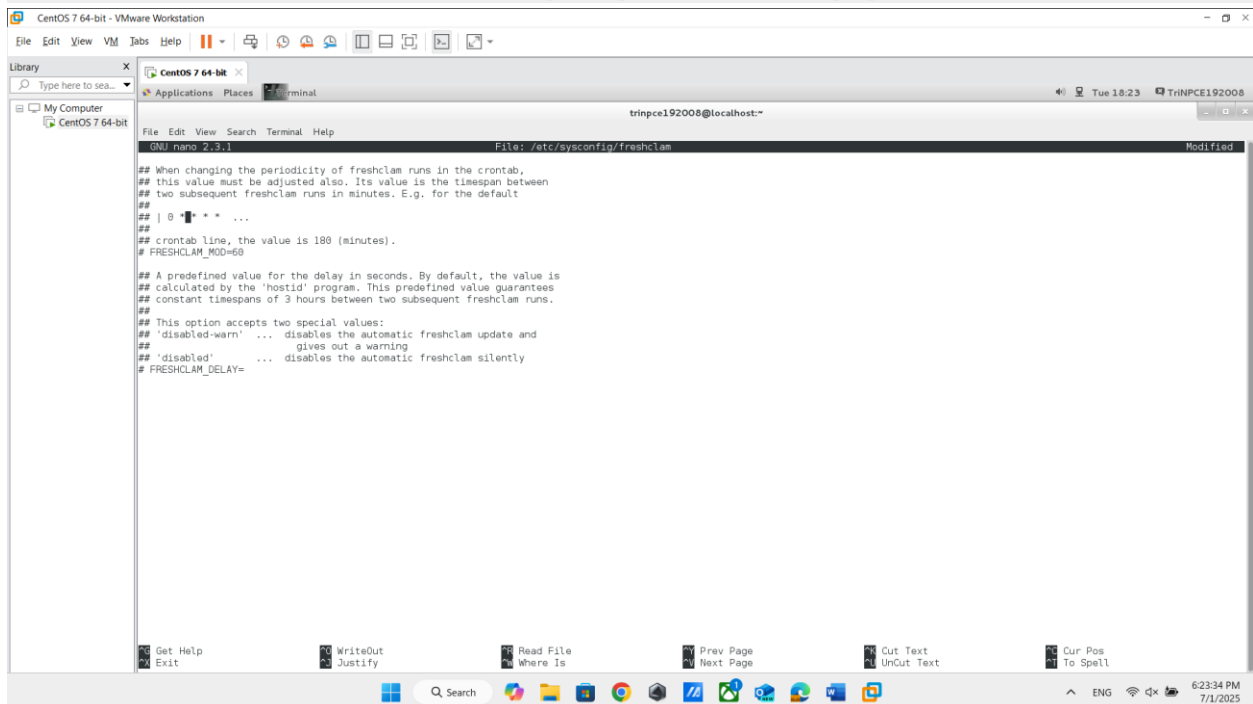
ClamAV:



The screenshot shows a terminal window titled 'CentOS 7 64-bit - VMware Workstation'. The user is editing the file `/etc/cron.d/clamav-update` using the `nano` editor. The file contains the following content:

```
GNU nano 2.3.1 File: /etc/cron.d/clamav-update
## Adjust this line...
MAILTO=root

## It is ok to execute it as root; freshclam drops privileges and becomes
## user 'clamupdate' as soon as possible
0 * * * root /usr/share/clamav/freshclam-sleep
```



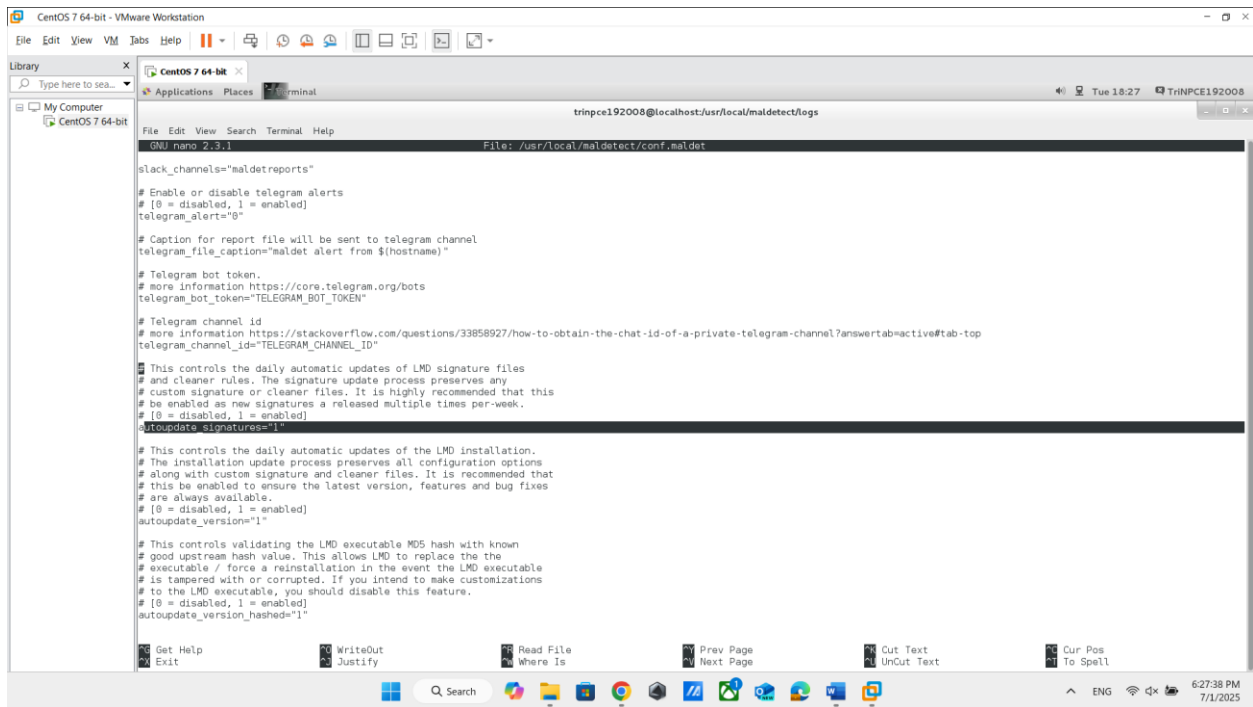
The screenshot shows a terminal window titled 'CentOS 7 64-bit - VMware Workstation'. The user is editing the file `/etc/sysconfig/freshclam` using the `nano` editor. The file contains the following content:

```
GNU nano 2.3.1 File: /etc/sysconfig/freshclam
## When changing the periodicity of freshclam runs in the crontab,
## this value must be adjusted also. Its value is the timespan between
## two subsequent freshclam runs in minutes. E.g. for the default
##
## | 0 * * * * ...
##
## crontab line, the value is 180 (minutes).
# FRESHCLAM_MOD=60

## A predefined value for the delay in seconds. By default, the value is
## calculated by the 'hostid' program. This predefined value guarantees
## constant timespans of 3 hours between two subsequent freshclam runs.
##
## This option accepts two special values:
## 'disabled-warn' ... disables the automatic freshclam update and
## 'disabled' ... gives out a warning
## 'disabled' ... disables the automatic freshclam silently
# FRESHCLAM_DELAY=
```

2 ảnh trên: bật tự động update cho clamAV mỗi giờ

Maldet:



```
File Edit View Search Terminal Help
GNU nano 2.3.1 File: /usr/local/maldet/conf/maldet

slack_channels="maldetreports"

# Enable or disable telegram alerts
# [0 = disabled, 1 = enabled]
telegram_alert="0"

# Caption for report file will be sent to telegram channel
telegram_file_caption="maldet alert from $(hostname)"

# Telegram bot token.
# more information https://core.telegram.org/bots
telegram_bot_token="TELEGRAM_BOT_TOKEN"

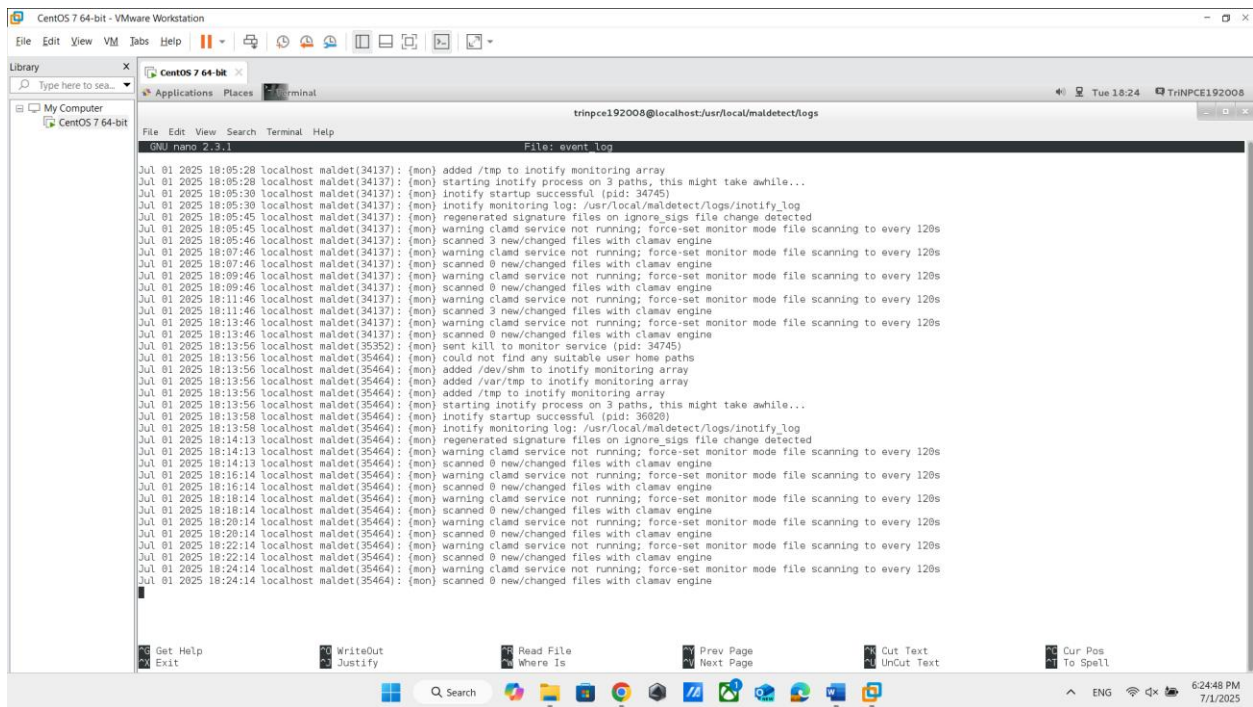
# Telegram channel id
# more information https://stackoverflow.com/questions/33858927/how-to-obtain-the-chat-id-of-a-private-telegram-channel?answertab=active#tab-top
telegram_channel_id="TELEGRAM_CHANNEL_ID"

# This controls the daily automatic updates of LMD signature files
# and cleaner rules. The signature update process preserves any
# custom signature or cleaner files. It is highly recommended that this
# be enabled as new signatures are released multiple times per-week.
# [0 = disabled, 1 = enabled]
autoupdate_signatures="1"

# This controls the daily automatic updates of the LMD installation.
# The installation update process preserves all configuration options
# along with custom signature and cleaner files. It is recommended that
# this be enabled to ensure the latest version, features and bug fixes
# are always available.
# [0 = disabled, 1 = enabled]
autoupdate_version="1"

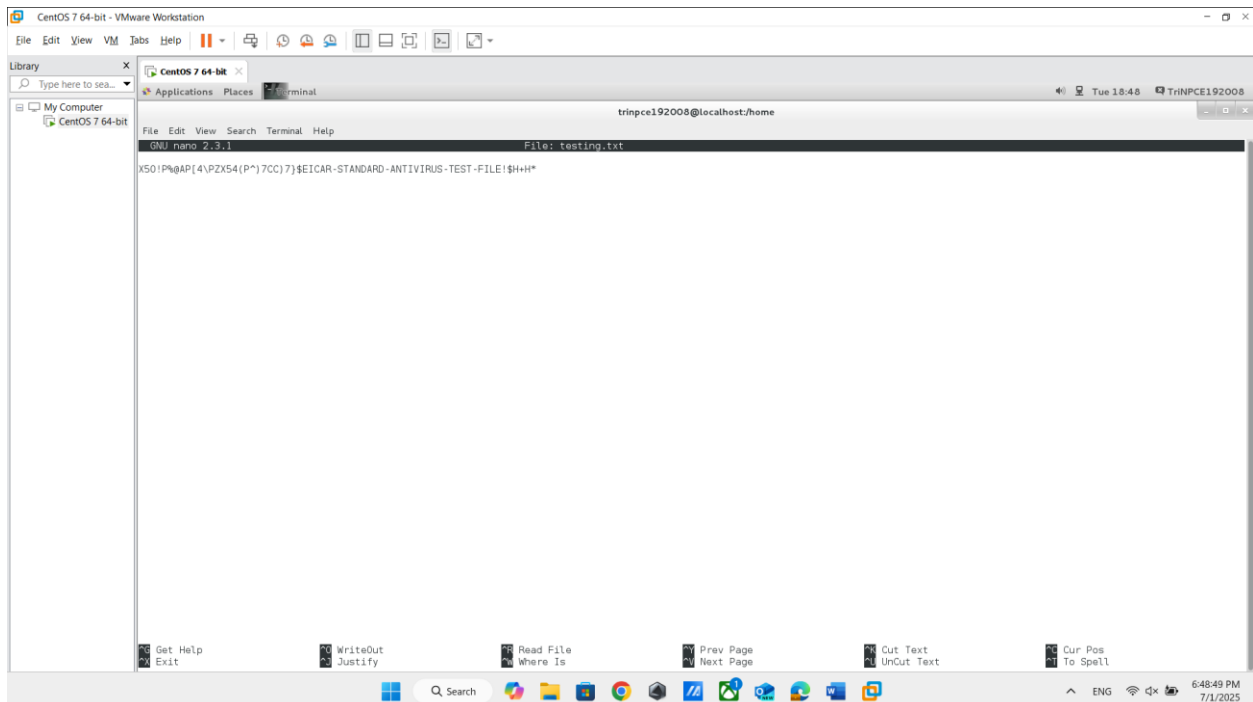
# This controls validating the LMD executable MD5 hash with known
# good upstream hash value. This allows LMD to replace the the
# executable / force a reinstallation in the event the LMD executable
# is tampered with or corrupted. If you intend to make customizations
# to the LMD executable, you should disable this feature.
# [0 = disabled, 1 = enabled]
autoupdate_version_hashed="1"

Get Help      WriteOut      Read File     Prev Page     Cut Text      Cur Pos
Exit          Justify       Where Is      Next Page     UnCut Text    To Spell
```

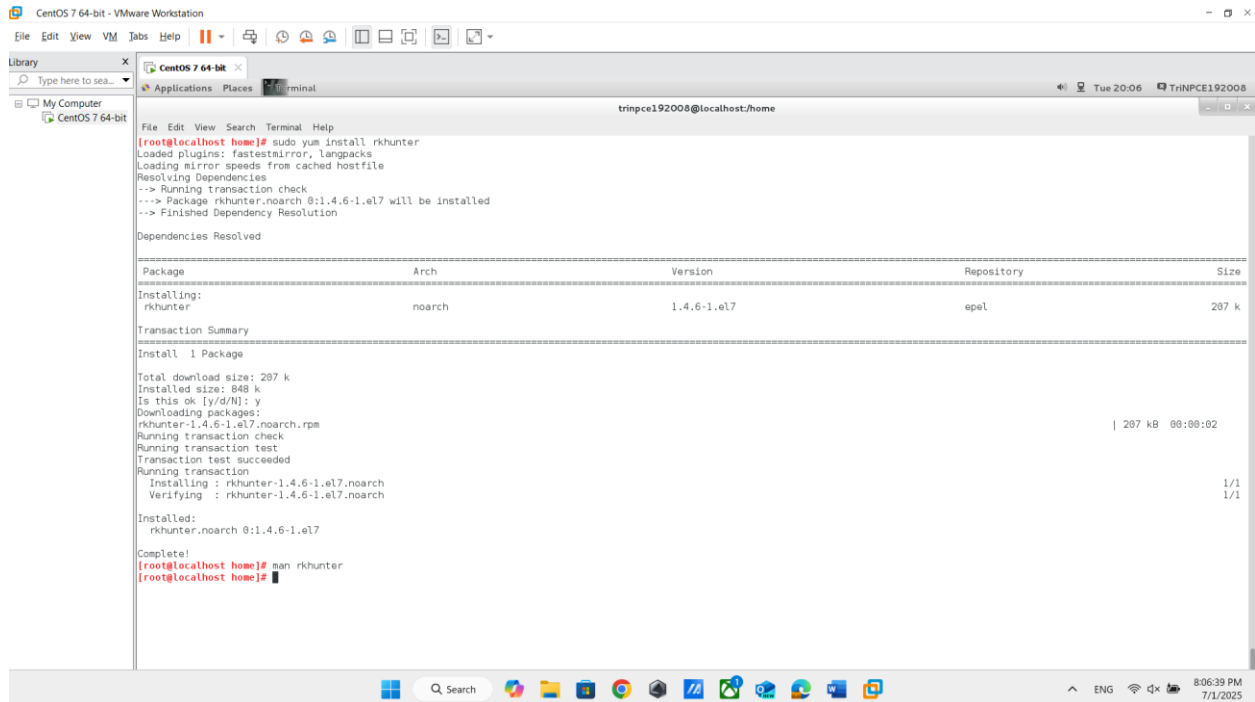


2 ảnh trên: bật tự động update cho maldet

Scanning with ClamAV and maldet



SELinux considerations:



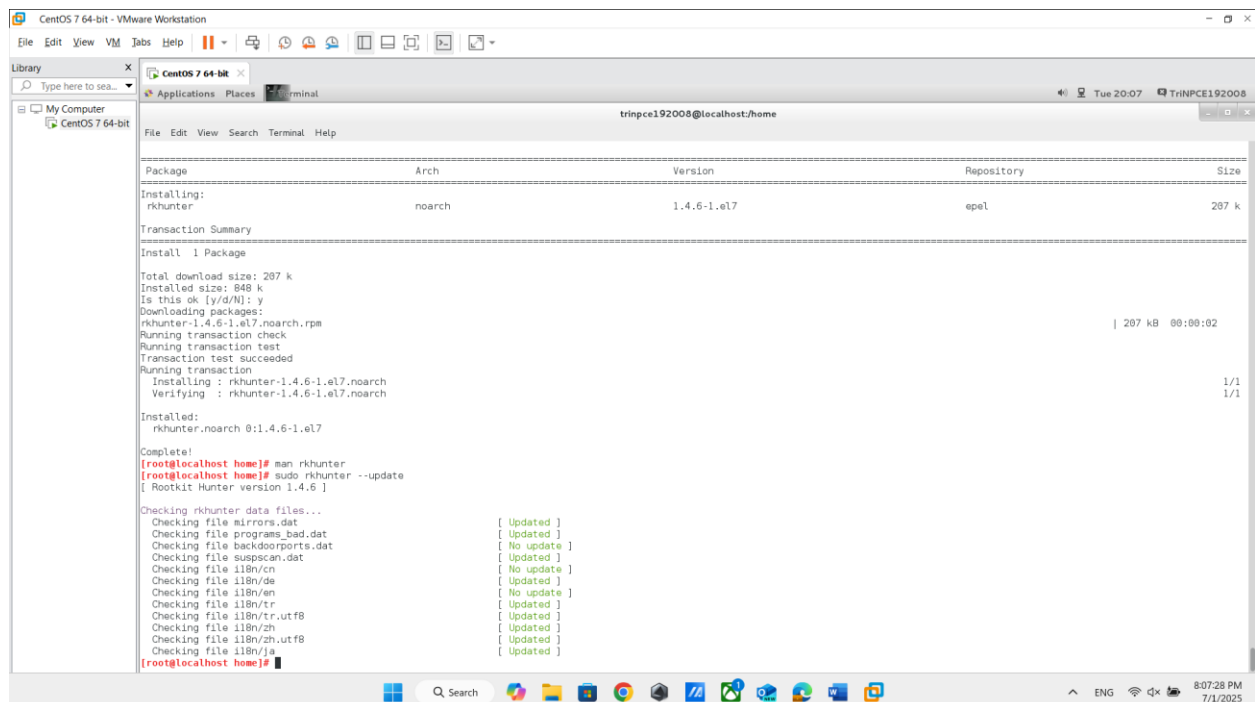
```
[root@localhost home]# sudo yum install rkhunter
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
Resolving Dependencies
--> Running transaction check
--> Package rkhunter.noarch 0:1.4.6-1.el7 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====================================================================================================================================
Package                               Arch                               Version                               Repository                               Size
=====================================================================================================================================
Installing:
rkhunter                             noarch                             1.4.6-1.el7                           epel                                     207 k
Transaction Summary
--
Install 1 Package
Total download size: 207 k
Installed size: 848 k
Is this ok [y/d/N]: y
Downloading packages:
rkhunter-1.4.6-1.el7.noarch.rpm
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : rkhunter-1.4.6-1.el7.noarch
Verifying : rkhunter-1.4.6-1.el7.noarch

Installed:
rkhunter.noarch 0:1.4.6-1.el7

Complete!
[root@localhost home]# man rkhunter
[root@localhost home]#
```



```
[root@localhost home]# sudo yum install rkhunter
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
Resolving Dependencies
--> Running transaction check
--> Package rkhunter.noarch 0:1.4.6-1.el7 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====================================================================================================================================
Package                               Arch                               Version                               Repository                               Size
=====================================================================================================================================
Installing:
rkhunter                             noarch                             1.4.6-1.el7                           epel                                     207 k
Transaction Summary
--
Install 1 Package
Total download size: 207 k
Installed size: 848 k
Is this ok [y/d/N]: y
Downloading packages:
rkhunter-1.4.6-1.el7.noarch.rpm
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : rkhunter-1.4.6-1.el7.noarch
Verifying : rkhunter-1.4.6-1.el7.noarch

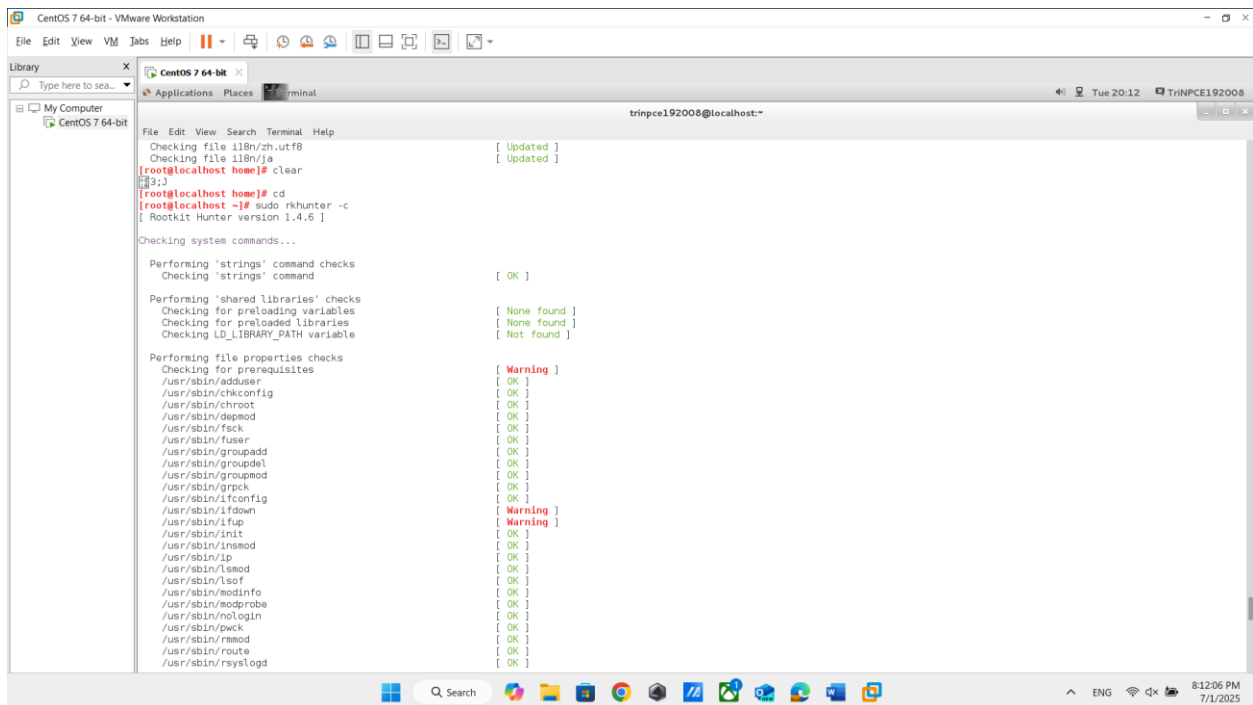
Installed:
rkhunter.noarch 0:1.4.6-1.el7

Complete!
[root@localhost home]# man rkhunter
[root@localhost home]# sudo rkhunter --update
[ Rootkit Hunter version 1.4.6 ]

Checking rkhunter data files...
Checking file mirrors.dat [ Updated ]
Checking file programs_bad.dat [ Updated ]
Checking file backdoorports.dat [ No update ]
Checking file suspcan.dat [ Updated ]
Checking file i18n/cn [ No update ]
Checking file i18n/de [ Updated ]
Checking file i18n/en [ No update ]
Checking file i18n/fr [ Updated ]
Checking file i18n/tr.utf8 [ Updated ]
Checking file i18n/zh [ Updated ]
Checking file i18n/zh.utf8 [ Updated ]
Checking file i18n/ja [ Updated ]
[root@localhost home]#
```

2 ảnh trên: cài đặt Rootkits với Rootkit Hunter và cập nhật chữ ký rkhunter – update

Scanning for rootkits:



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
CentOS 7 64-bit
Applications Places
Terminal
trinpce192008@localhost:~

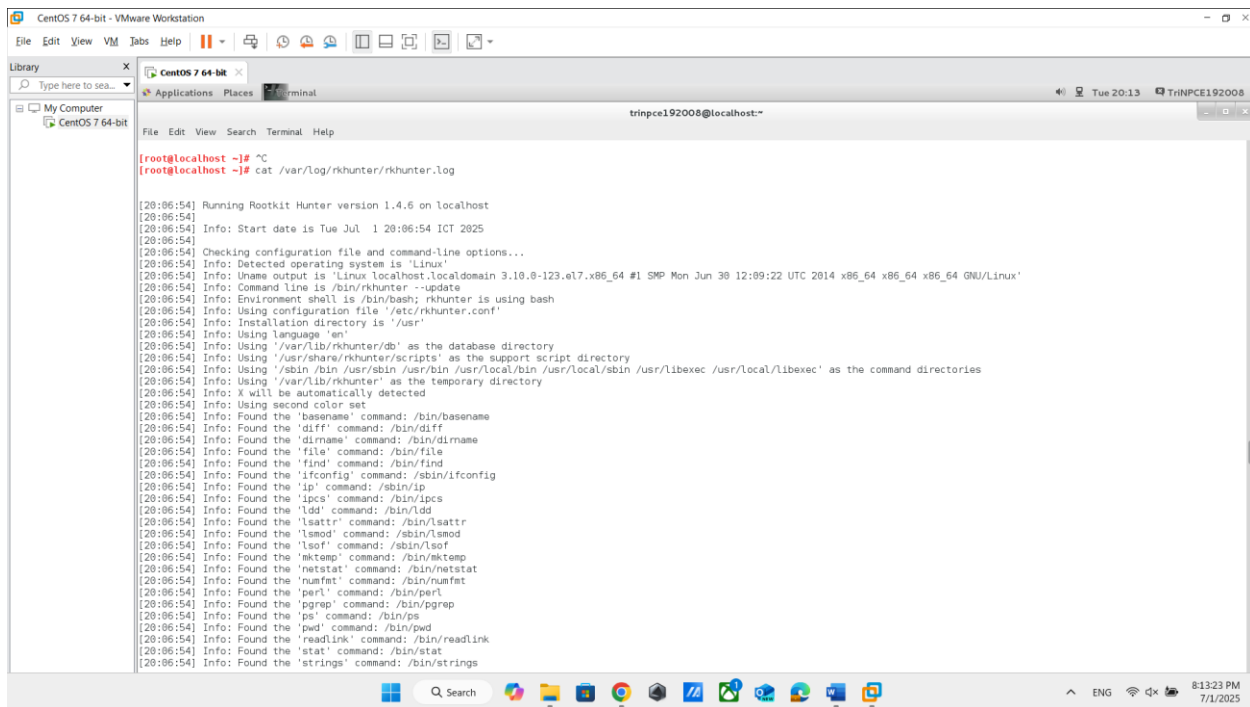
File Edit View Search Terminal Help
Checking file i18n/zh.utf8 [ Updated ]
Checking file i18n/ja [ Updated ]
[root@localhost home]# clear
[3;3;]
[root@localhost home]# cd
[root@localhost ~]# sudo rkhunter -c
[ Rootkit Hunter version 1.4.6 ]

Checking system commands...

Performing 'strings' command checks
Checking 'strings' command [ OK ]

Performing 'shared libraries' checks
Checking for preloading variables [ None found ]
Checking for preloaded libraries [ None found ]
Checking LD_LIBRARY_PATH variable [ Not found ]

Performing file properties checks
Checking for prerequisites [ Warning ]
/usr/sbin/adduser [ OK ]
/usr/sbin/chkconfig [ OK ]
/usr/sbin/chroot [ OK ]
/usr/sbin/depmod [ OK ]
/usr/sbin/fsck [ OK ]
/usr/sbin/fuser [ OK ]
/usr/sbin/groupadd [ OK ]
/usr/sbin/groupdel [ OK ]
/usr/sbin/groupmod [ OK ]
/usr/sbin/grpck [ OK ]
/usr/sbin/ifconfig [ OK ]
/usr/sbin/ifdown [ Warning ]
/usr/sbin/ifup [ Warning ]
/usr/sbin/init [ OK ]
/usr/sbin/insmod [ OK ]
/usr/sbin/ip [ OK ]
/usr/sbin/lsmmod [ OK ]
/usr/sbin/lsof [ OK ]
/usr/sbin/modinfo [ OK ]
/usr/sbin/modprobe [ OK ]
/usr/sbin/nologin [ OK ]
/usr/sbin/pwck [ OK ]
/usr/sbin/rmmod [ OK ]
/usr/sbin/route [ OK ]
/usr/sbin/rsyslogd [ OK ]
```

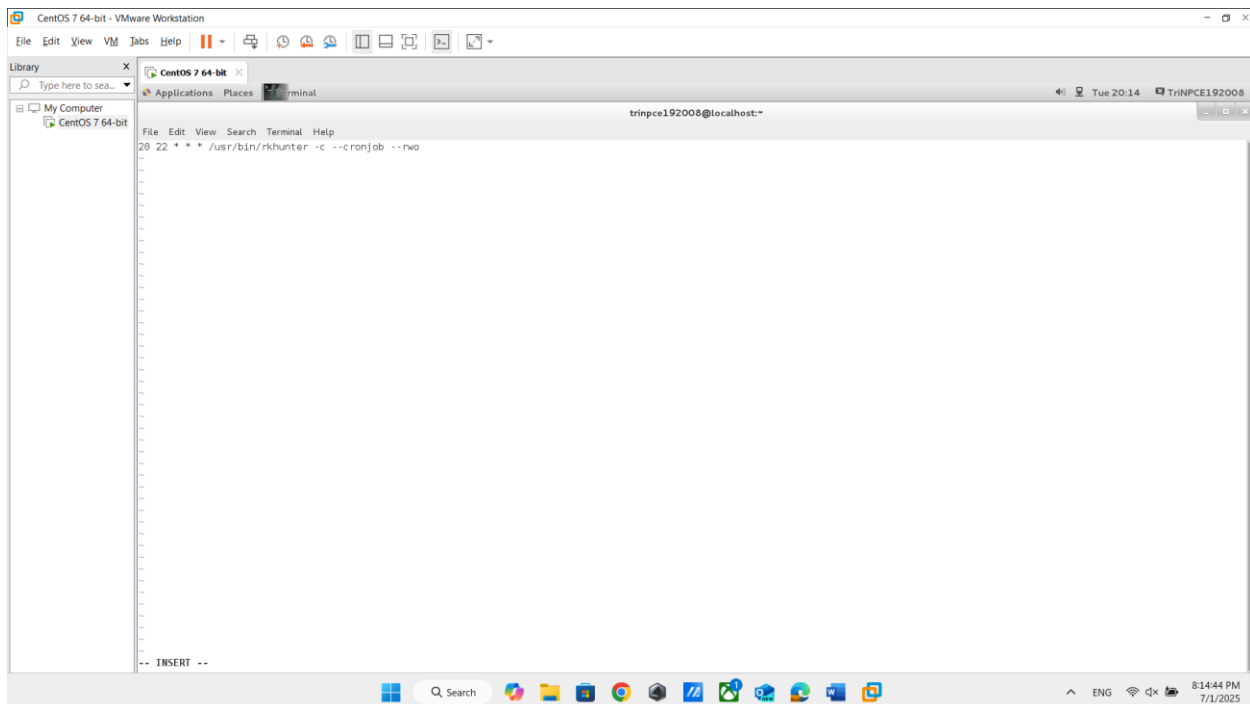


```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost:~

[root@localhost ~]# ^C
[root@localhost ~]# cat /var/log/rkhunter/rkhunter.log

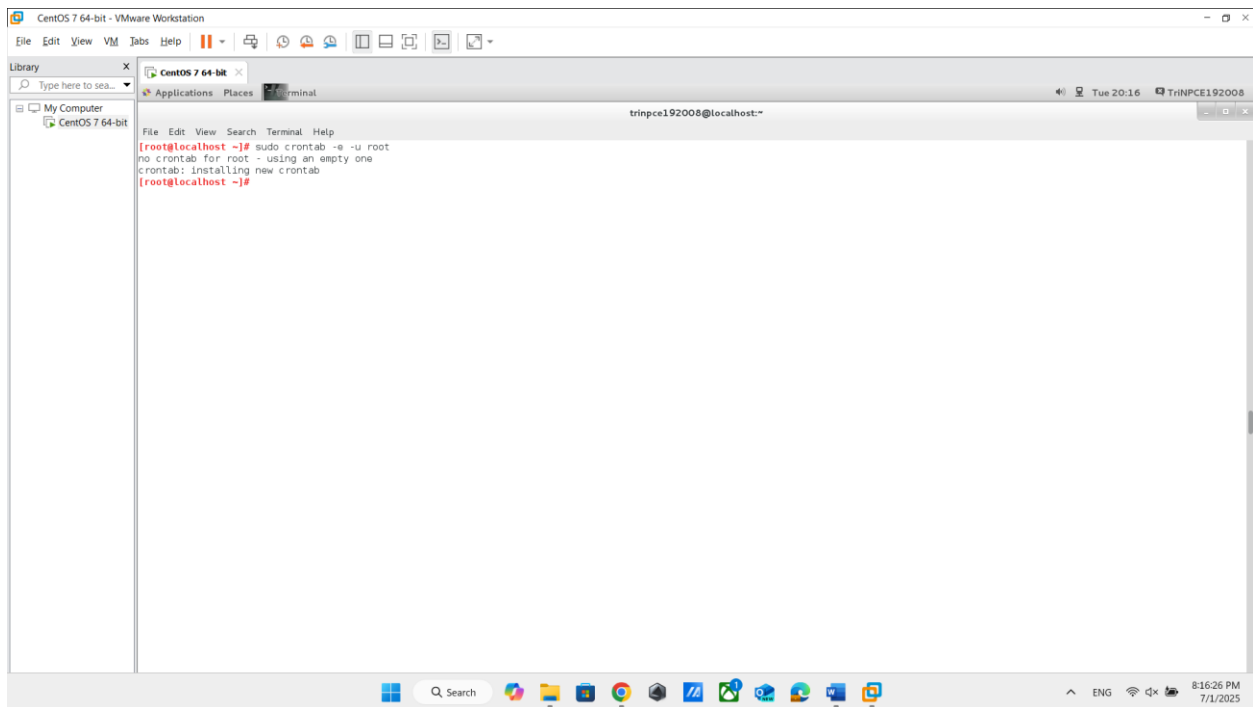
[20:06:54] Running Rootkit Hunter version 1.4.6 on localhost
[20:06:54] Info: Start date is Tue Jul 1 20:06:54 ICT 2025
[20:06:54] Info: Checking configuration file and command-line options...
[20:06:54] Info: Detected operating system is 'Linux'
[20:06:54] Info: Uname output is 'Linux localhost.localdomain 3.10.0-123.el7.x86_64 #1 SMP Mon Jun 30 12:09:22 UTC 2014 x86_64 x86_64 GNU/Linux'
[20:06:54] Info: Command line is '/bin/rkhunter --update'
[20:06:54] Info: Environment shell is '/bin/bash'; rkhunter is using bash
[20:06:54] Info: Using configuration file '/etc/rkhunter.conf'
[20:06:54] Info: Installation directory is '/usr'
[20:06:54] Info: Using language 'en'
[20:06:54] Info: Using '/var/lib/rkhunter/db' as the database directory
[20:06:54] Info: Using '/usr/share/rkhunter/scripts' as the support script directory
[20:06:54] Info: Using '/sbin:/bin:/usr/sbin:/usr/bin:/usr/local/bin:/usr/local/sbin:/usr/libexec:/usr/local/libexec' as the command directories
[20:06:54] Info: Using '/var/lib/rkhunter' as the temporary directory
[20:06:54] Info: X will be automatically detected
[20:06:54] Info: Using second color set
[20:06:54] Info: Found the 'basename' command: /bin/basename
[20:06:54] Info: Found the 'diff' command: /bin/diff
[20:06:54] Info: Found the 'dirname' command: /bin/dirmame
[20:06:54] Info: Found the 'file' command: /bin/file
[20:06:54] Info: Found the 'find' command: /bin/find
[20:06:54] Info: Found the 'ifconfig' command: /sbin/ifconfig
[20:06:54] Info: Found the 'ip' command: /sbin/ip
[20:06:54] Info: Found the 'lpc' command: /bin/lpc
[20:06:54] Info: Found the 'ldd' command: /bin/ldd
[20:06:54] Info: Found the 'lsattr' command: /bin/lsattr
[20:06:54] Info: Found the 'lsmod' command: /sbin/lsmod
[20:06:54] Info: Found the 'lsuf' command: /bin/lsuf
[20:06:54] Info: Found the 'mktemp' command: /bin/mktemp
[20:06:54] Info: Found the 'netstat' command: /bin/netstat
[20:06:54] Info: Found the 'numfmt' command: /bin/numfmt
[20:06:54] Info: Found the 'perl' command: /bin/perl
[20:06:54] Info: Found the 'pgrep' command: /bin/pgrep
[20:06:54] Info: Found the 'ps' command: /bin/ps
[20:06:54] Info: Found the 'pwd' command: /bin/pwd
[20:06:54] Info: Found the 'readlink' command: /bin/readlink
[20:06:54] Info: Found the 'stat' command: /bin/stat
[20:06:54] Info: Found the 'strings' command: /bin/strings
```

2 ảnh trên: scan với rootkits bằng lệnh rkhunter -c, file log sẽ được lưu vào /var/log/rkhunter/rkhunter.log



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
CentOS 7 64-bit
Applications Places Terminal
trinpce192008@localhost:~

20 22 * * * /usr/bin/rkhunter -c --cronjob --two
-- INSERT --
```



2 ảnh trên: Mở crontab cho root bằng lệnh `sudo crontab -e -u root` và thêm dòng `20 22 * * * /usr/bin/rkhunter -c --cronjob -rwo`

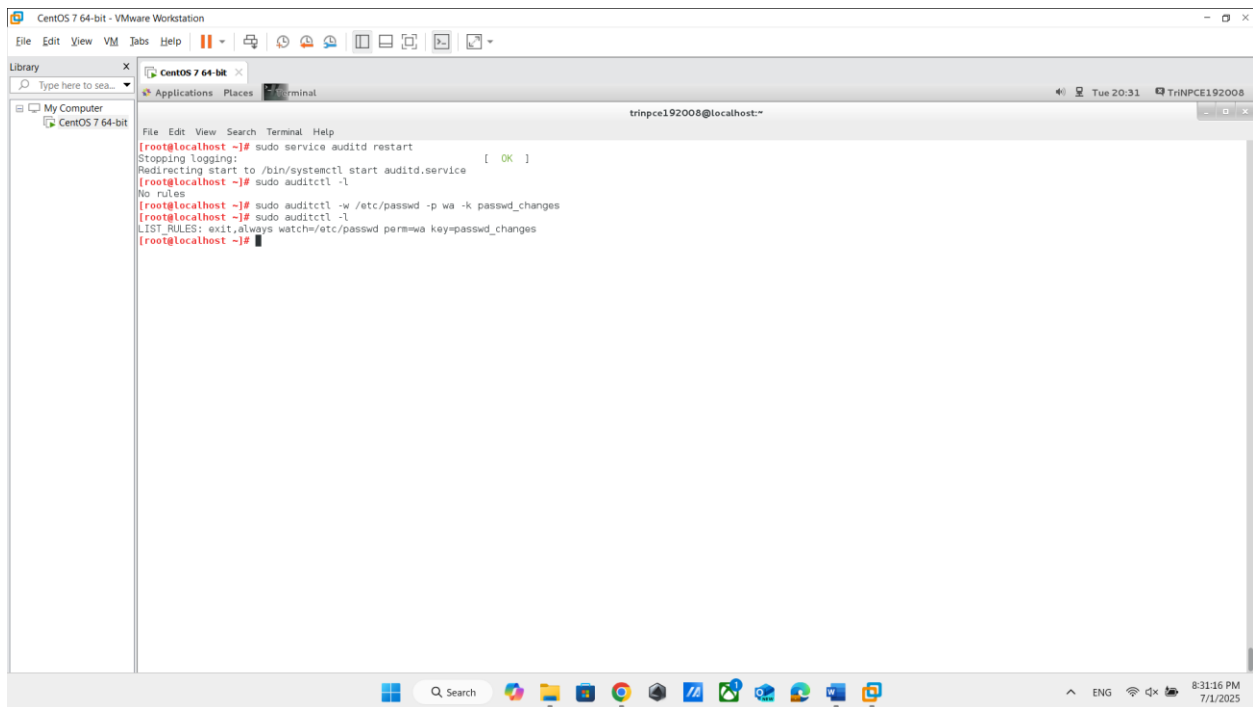
Ý nghĩa:

Chạy lúc 22:20 hằng đêm

--cronjob: chạy không cần tương tác

--rwo: chỉ hiện cảnh báo (warning)

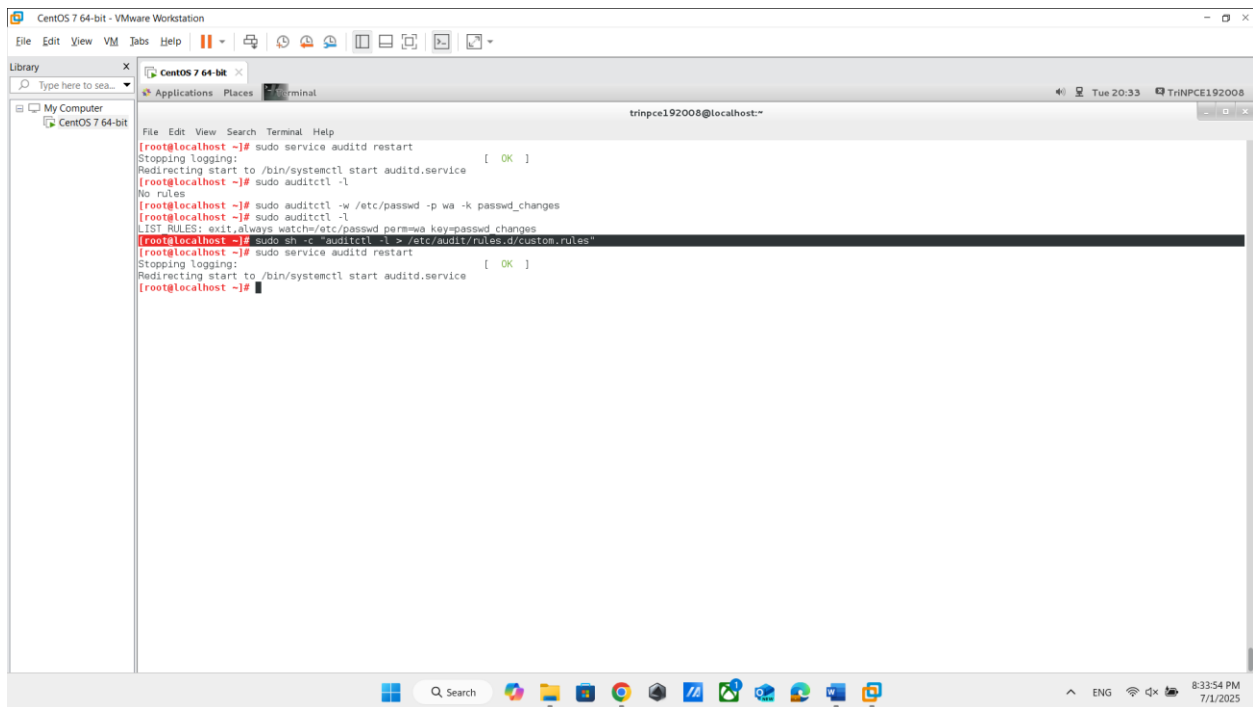
Controlling the auditd daemon & Creating audit rules



ảnh trên: restart auditd trên Centos 7, kiểm tra danh sách các rule đã đặt, tạo 1 rule giám sát file /etc/passwd

Giải thích:

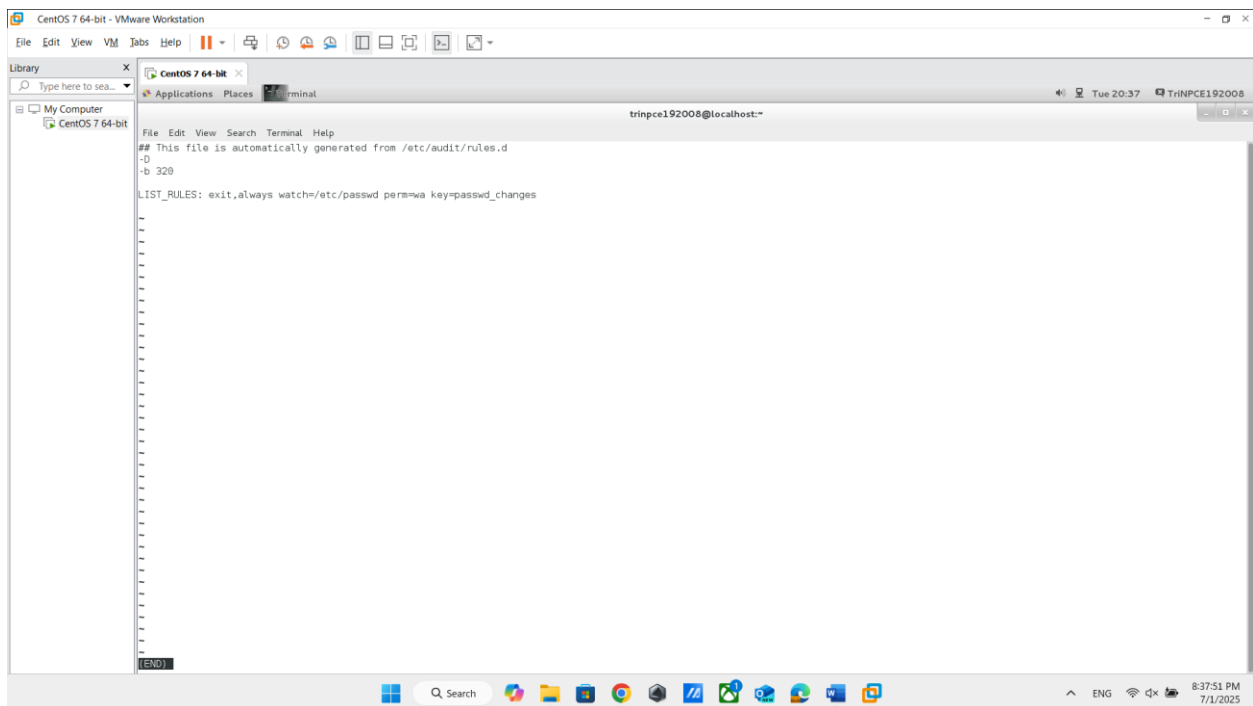
- **-w:** theo dõi đường dẫn
 - **-p wa:** theo dõi hành vi ghi (write) và thay đổi thuộc tính (attribute change)
 - **-k passwd_changes:** gắn nhãn để tra cứu
-



```
trinpce192008@localhost:~$ sudo service auditd restart
Stopping logging:
Redirecting start to /bin/systemctl start auditd.service [ OK ]
trinpce192008@localhost:~$ sudo auditctl -l
No rules
trinpce192008@localhost:~$ sudo auditctl -w /etc/passwd -p wa -k passwd_changes
LIST RULES: exit,always watch=/etc/passwd perm=wa key=passwd_changes
trinpce192008@localhost:~$ sudo sh -c "auditctl -l > /etc/audit/rules.d/custom.rules"
trinpce192008@localhost:~$ sudo service auditd restart
Stopping logging:
Redirecting start to /bin/systemctl start auditd.service [ OK ]
trinpce192008@localhost:~$
```

ảnh trên: vì luật sẽ mất khi reboot ta sẽ lưu rule vừa tạo vào file `/etc/audit/rules.d/custom.rules`

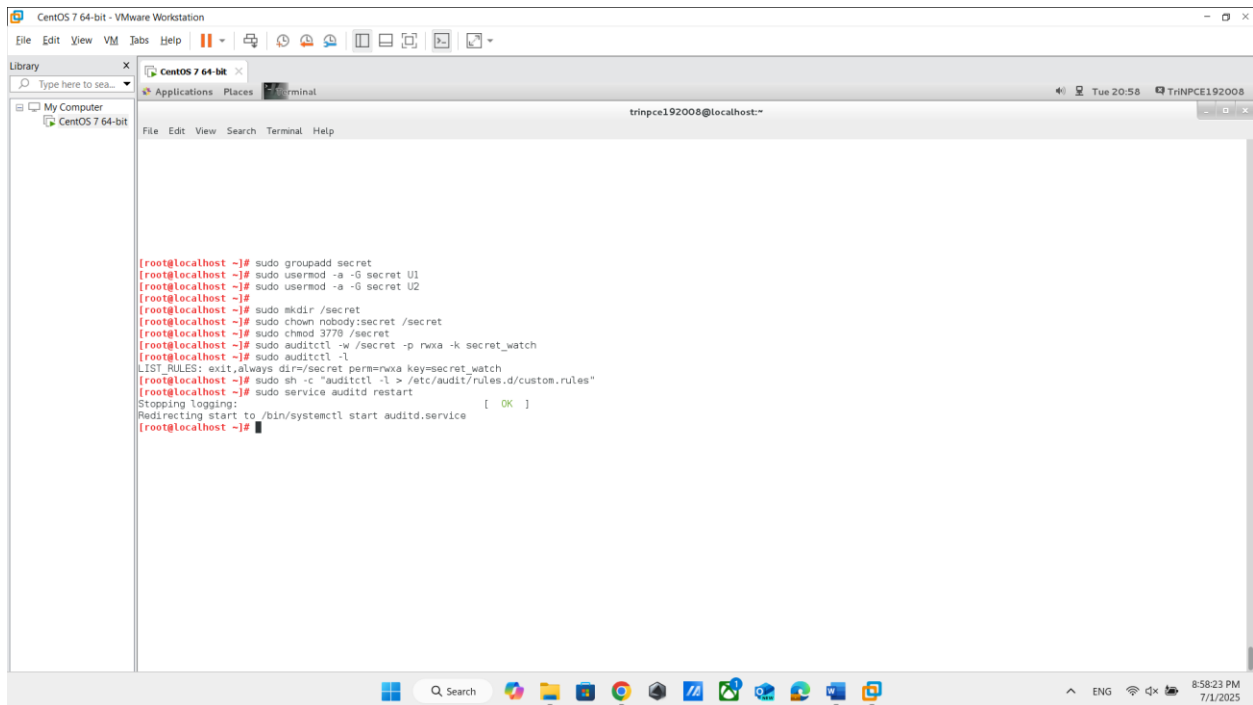
- `auditctl -l`: in ra tất cả luật đang có.
- `> /etc/audit/rules.d/custom.rules`: ghi ra file mới để lưu vĩnh viễn.



```
trinpce192008@localhost:~$ cat /etc/audit/rules.d/custom.rules
## This file is automatically generated from /etc/audit/rules.d
-d
-b 320
LIST_RULES: exit,always watch=/etc/passwd perm=wa key=passwd_changes
```

ảnh trên: Sau khi reboot hoặc restart auditd, file /etc/audit/audit.rules sẽ chứa luật mới

Auditing a directory:



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
CentOS 7 64-bit
Applications Places
trinpce192008@localhost:~
File Edit View Search Terminal Help

[root@localhost ~]# sudo groupadd secret
[root@localhost ~]# sudo usermod -a -G secret U1
[root@localhost ~]# sudo usermod -a -G secret U2
[root@localhost ~]#
[root@localhost ~]# sudo mkdir /secret
[root@localhost ~]# sudo chown nobody:secret /secret
[root@localhost ~]# sudo chmod 3779 /secret
[root@localhost ~]# sudo auditctl -w /secret -p rwx -k secret_watch
[root@localhost ~]# sudo auditctl -l
LIST RULES: exit,always dir=/secret perm=rwx key=secret_watch
[root@localhost ~]# sudo sh -c "auditctl -l > /etc/audit/rules.d/custom.rules"
[root@localhost ~]# sudo service auditd restart
Stopping logging: [ OK ]
Redirecting start to /bin/systemctl start auditd.service
[root@localhost ~]#
```

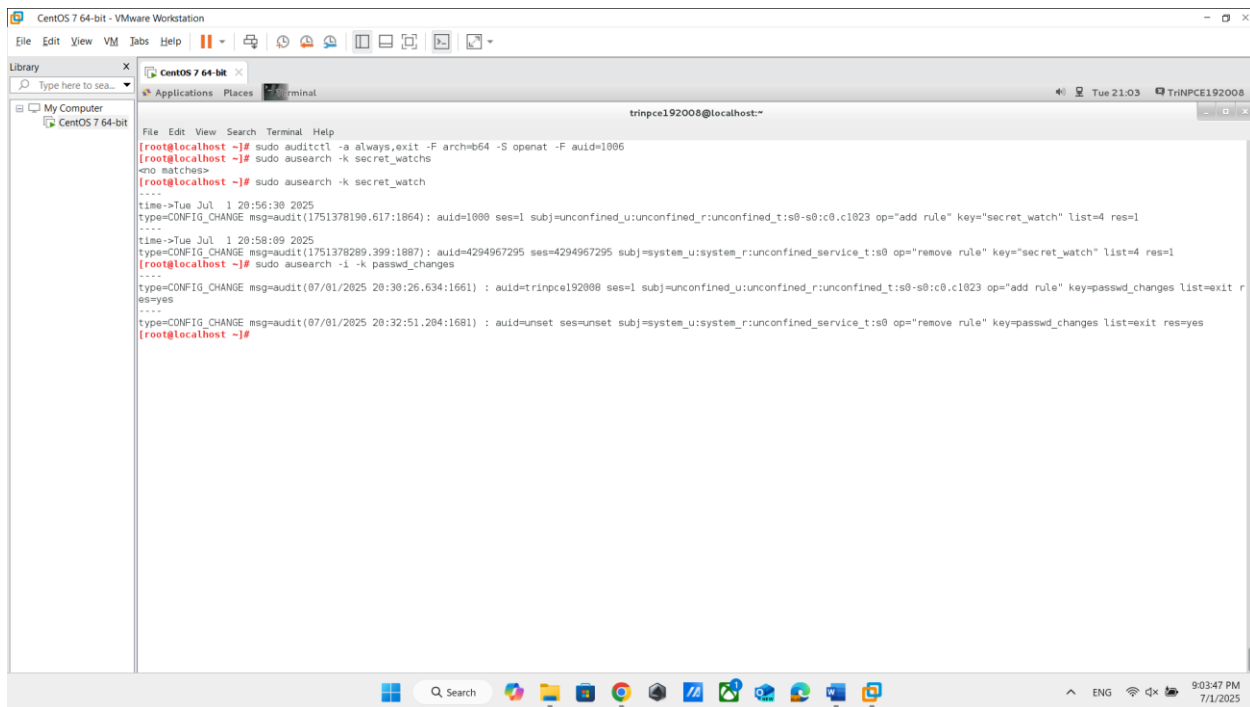
ảnh trên: Khởi tạo thư mục và nhóm cần giám sát giả sử muốn giám sát thư mục /secret, thêm luật audit cho thư mục

Ý nghĩa:

- w: theo dõi một đường dẫn.
- p rwx: theo dõi hành động đọc (r), ghi (w), thay đổi quyền (a), thực thi (x).
- k secret_watch: gắn nhãn (key) để dễ tìm kiếm log sau này.

Lưu luật lại để tránh mất khi reboot

Auditing system calls and Using ausearch and aureport:



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Help
Library
Type here to search
Applications Places Terminal
trinpce192008@localhost:~$
[root@localhost ~]# sudo auditctl -a always,exit -F arch=b64 -S openat -F auid=1006
[root@localhost ~]# sudo ausearch -k secret_watch
<no matches>
[root@localhost ~]# sudo ausearch -k secret_watch
----
time-->Tue Jul 1 20:56:30 2025
type=CONFIG_CHANGE msg=audit(1751378190.617:1864) : auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 op="add rule" key="secret_watch" list=4 res=1
----
time-->Tue Jul 1 20:58:09 2025
type=CONFIG_CHANGE msg=audit(1751378289.399:1887) : auid=4294967295 ses=4294967295 subj=system_u:system_r:unconfined_service_t:s0 op="remove rule" key="secret_watch" list=4 res=1
[root@localhost ~]# sudo ausearch -i -k passwd_changes
----
type=CONFIG_CHANGE msg=audit(07/01/2025 20:30:26.634:1661) : auid=trinpce192008 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 op="add rule" key=passwd_changes list=exit r
es=yes
----
type=CONFIG_CHANGE msg=audit(07/01/2025 20:32:51.204:1681) : auid=unset ses=unset subj=system_u:system_r:unconfined_service_t:s0 op="remove rule" key=passwd_changes list=exit res=yes
[root@localhost ~]#
```

ảnh trên: muốn giám sát user có uid=1006 khi dùng lệnh openat (mở file)
sudo auditctl -a always,exit -F arch=b64 -S openat -F auid=1006

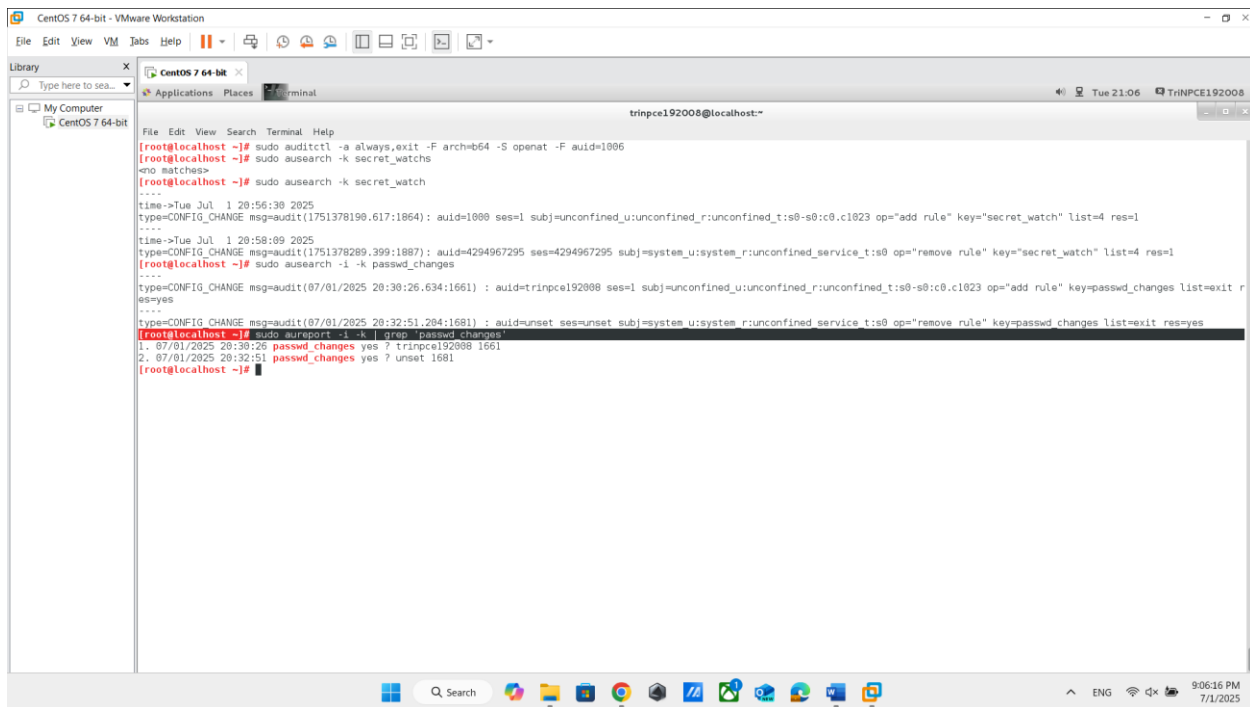
Giải thích:

- **-a always,exit:** luôn tạo log khi system call kết thúc.
- **-F arch=b64:** kiến trúc CPU 64-bit.
- **-S openat:** gọi syscall mở file.
- **-F auid=1006:** lọc theo UID người dùng (đăng nhập bằng uid=1006).

Để dễ dàng tìm kiếm ví dụ tìm các hành động liên quan đến key secret_watch hoặc theo key passwd_changes ta dùng

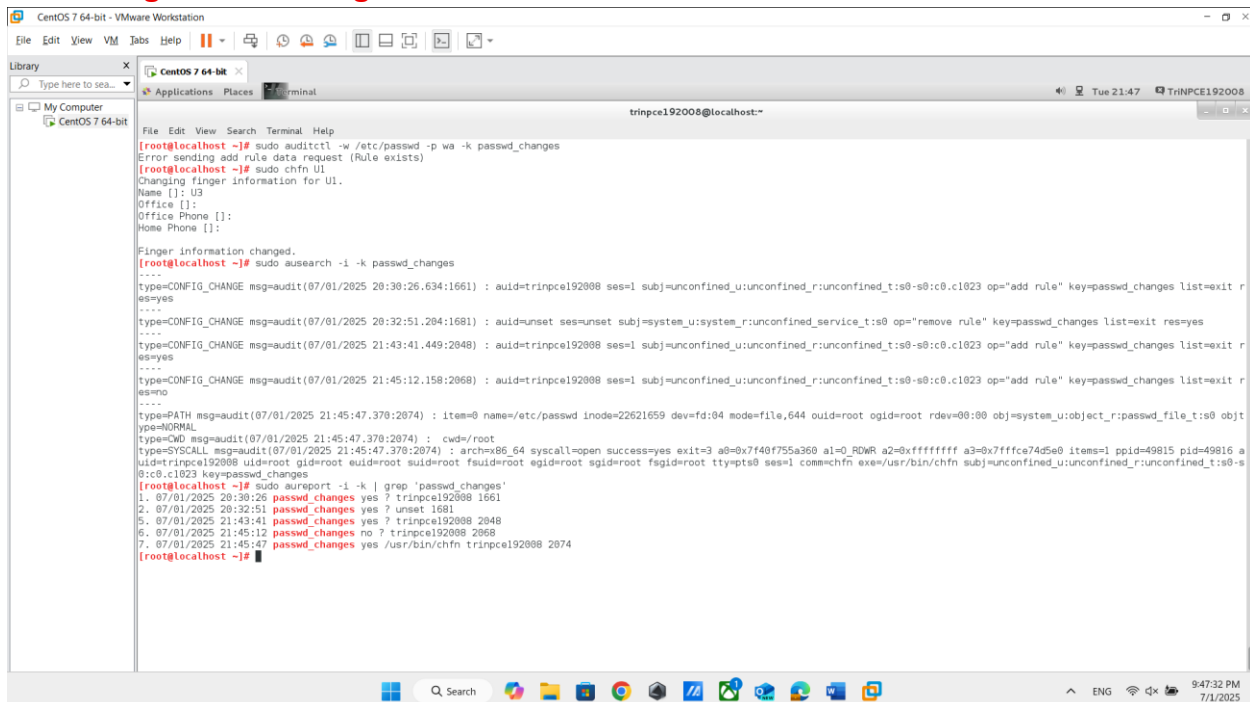
sudo ausearch -k secretcats_watch

sudo ausearch -i -k passwd_changes



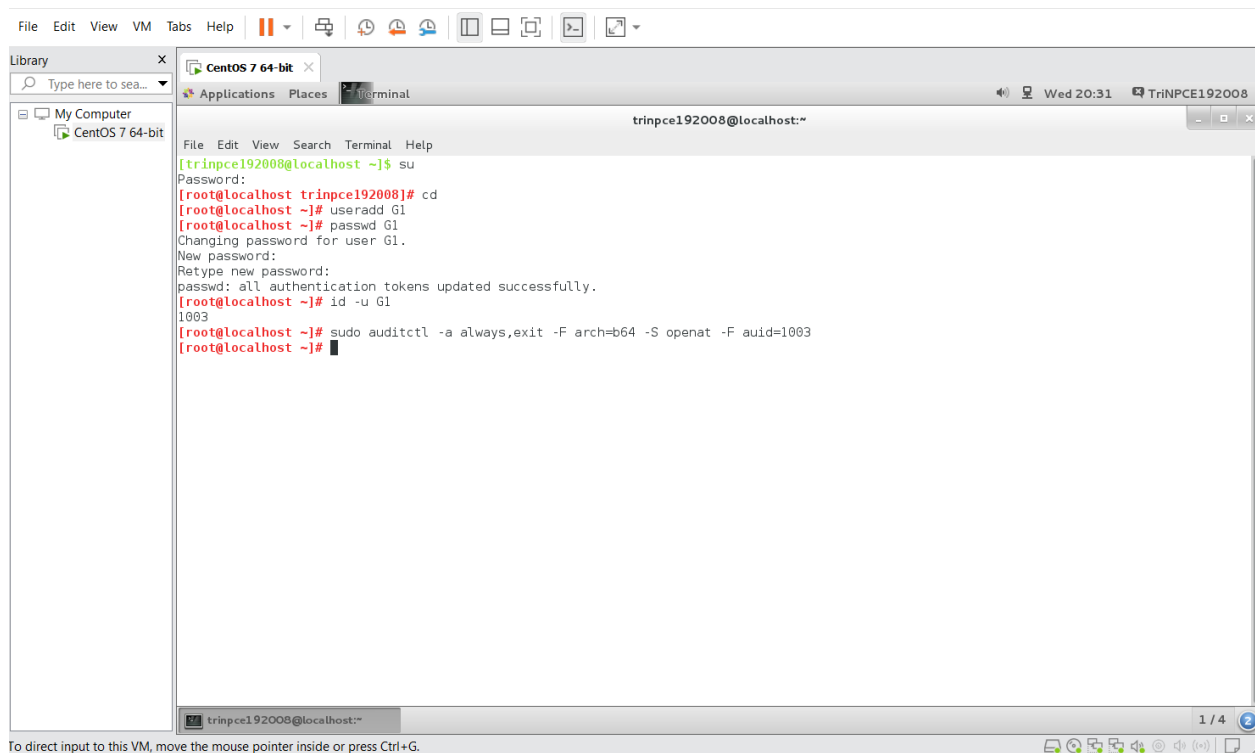
ảnh trên: sử dụng aureport liệt kê nhanh các hành động theo key để dễ quản lý thay vì phải xem file log

Searching for file change alerts:



ảnh trên: Thực hiện thay đổi để kiểm tra (sử dụng chfn để đổi thông tin người dùng), Tìm kiếm các sự kiện bằng ausearch, Tổng hợp sự kiện bằng aureport ta thấy : yes /usr/bin/chfn trinpce192008 2074 là hành động làm thay đổi /etc/passwd.

Searching for system call rule violations:

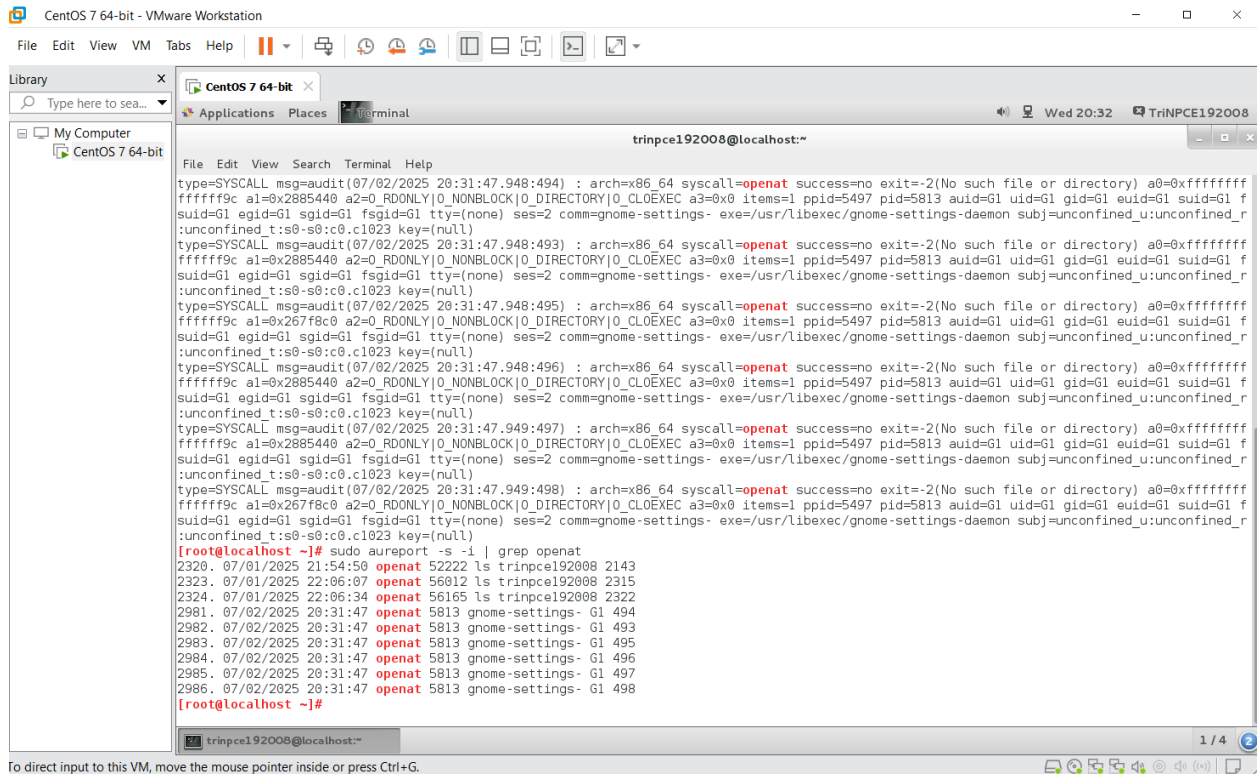


The screenshot shows a terminal window titled 'CentOS 7 64-bit' with a 'Terminal' tab selected. The user 'trinpce192008@localhost' is logged in. The terminal output shows the following commands and their results:

```
[trinpce192008@localhost ~]$ su
Password:
[root@localhost trinpce192008]# cd
[root@localhost ~]# useradd G1
[root@localhost ~]# passwd G1
Changing password for user G1.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# id -u G1
1003
[root@localhost ~]# sudo auditctl -a always,exit -F arch=b64 -S openat -F auid=1003
[root@localhost ~]#
```

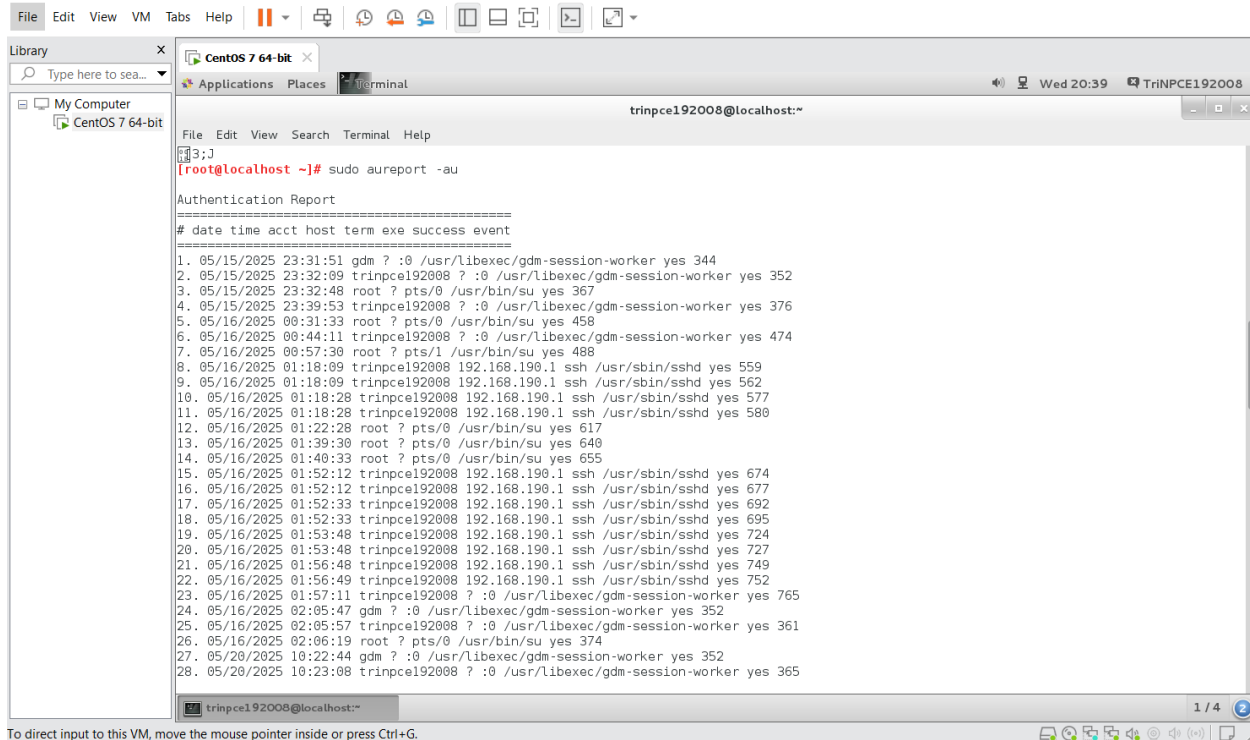
ảnh trên: thiết lập quy tắc audit Giải thích:

- a always,exit Ghi log khi system call kết thúc
- F arch=b64 Áp dụng cho hệ thống 64-bit
- S openat Ghi lại các lời gọi hệ thống (syscall) mở hoặc tạo file
- F auid=1003 Áp dụng cho user có UID 1003 (G1)

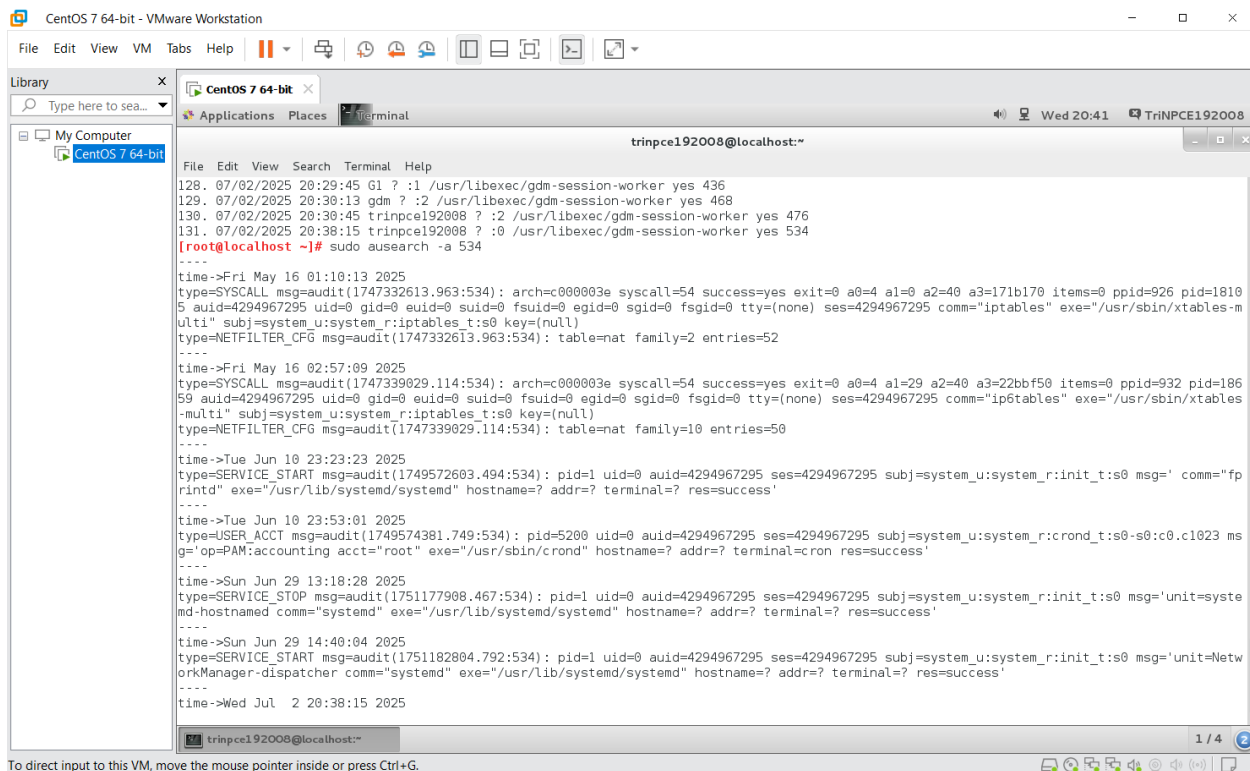


ảnh trên: Kiểm tra log ghi nhận hành vi của G1

Generating authentication reports:

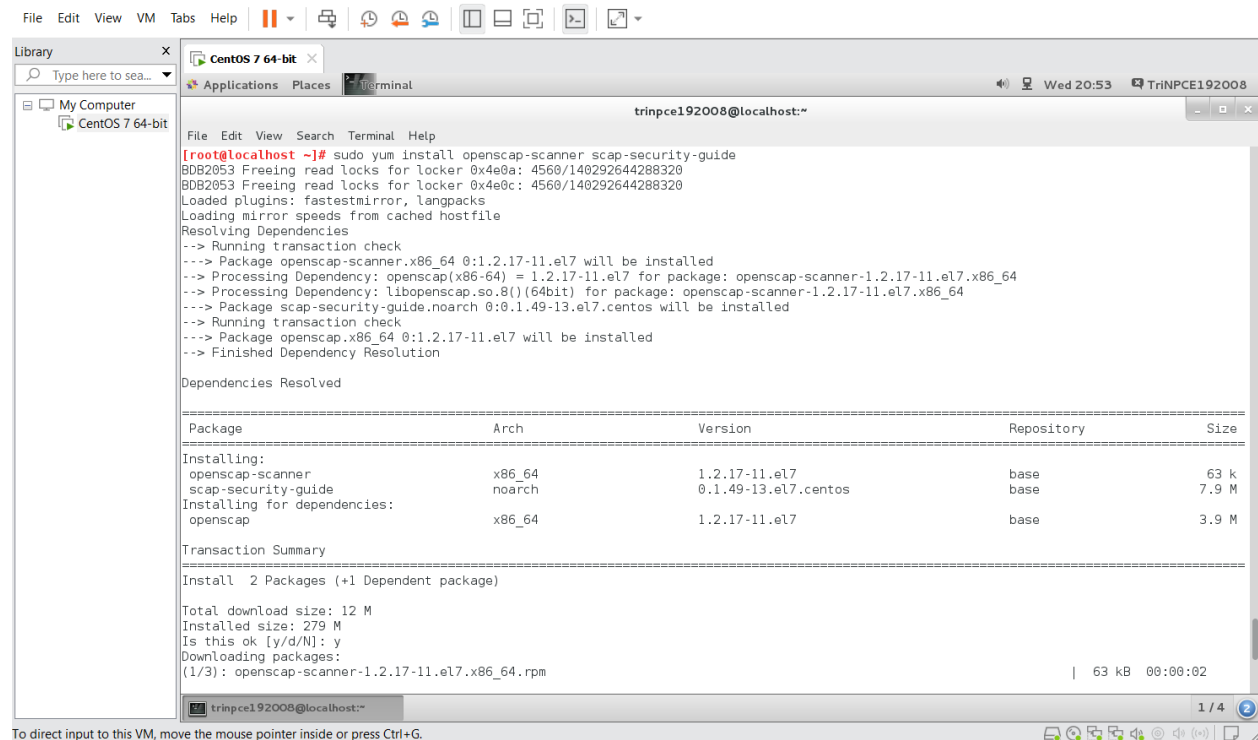


ảnh trên: Xem lịch sử đăng nhập và ID của các event



ảnh trên: `sudo ausearch -a` để xem chi tiết, Dùng `ausearch -a <event_id>` để xem chi tiết và lấy auid. auid hữu ích trong audit log vì cho biết ai thật sự login vào hệ thống, kể cả khi họ dùng `sudo` để chạy lệnh với quyền khác.

Installing OpenSCAP



```
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
CentOS 7 64-bit
Applications Places Terminal
trinpc192008@localhost:~$ sudo yum install openscap-scanner scap-security-guide
BDB2053 Freeing read locks for locker 0x4e0a: 4560/140292644288320
BDB2053 Freeing read locks for locker 0x4e0c: 4560/140292644288320
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
Resolving Dependencies
--> Running transaction check
--> Package openscap-scanner.x86_64 0:1.2.17-11.el7 will be installed
--> Processing Dependency: openscap(x86-64) = 1.2.17-11.el7 for package: openscap-scanner-1.2.17-11.el7.x86_64
--> Processing Dependency: libopenscap.so.8()(64bit) for package: openscap-scanner-1.2.17-11.el7.x86_64
--> Package scap-security-guide.noarch 0:0.1.49-13.el7.centos will be installed
--> Running transaction check
--> Package openscap.x86_64 0:1.2.17-11.el7 will be installed
--> Finished Dependency Resolution

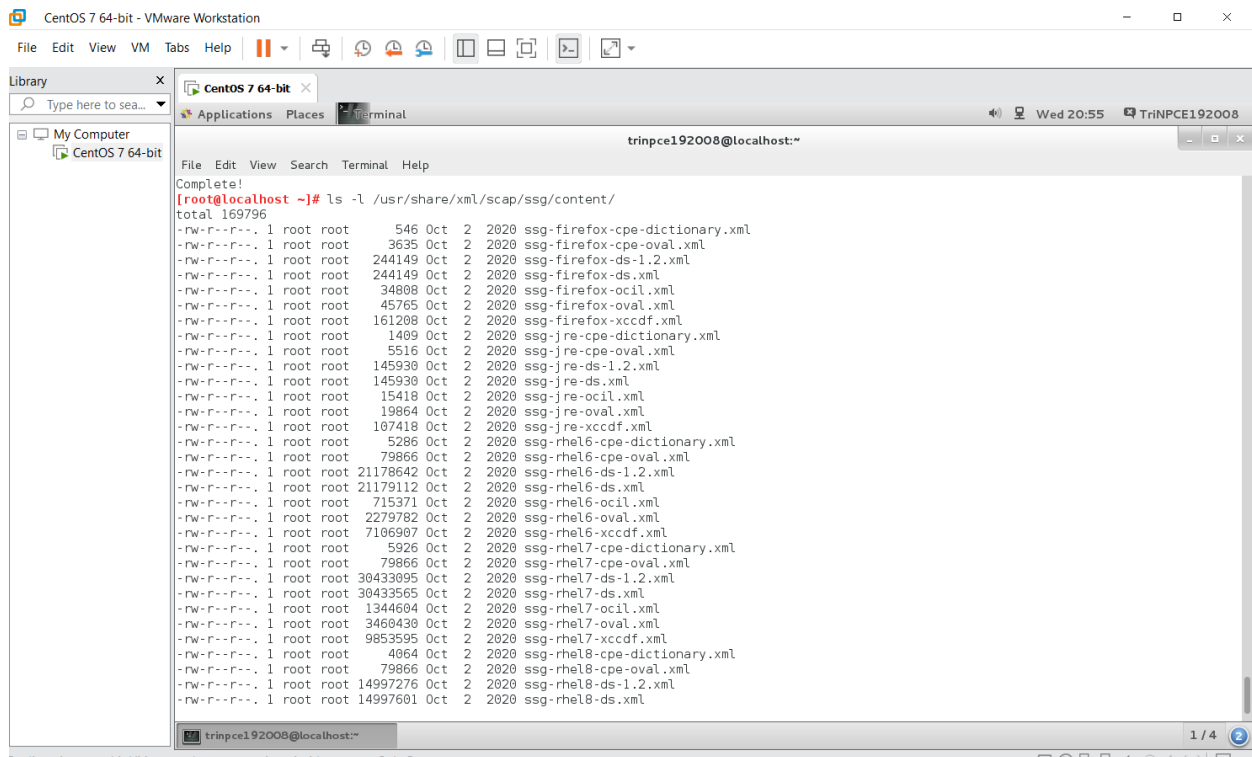
Dependencies Resolved

=====
Package Arch Version Repository Size
=====
Installing:
openscap-scanner x86_64 1.2.17-11.el7 base 63 k
scap-security-guide noarch 0.1.49-13.el7.centos base 7.9 M
Installing for dependencies:
openscap x86_64 1.2.17-11.el7 base 3.9 M
=====
Transaction Summary
=====
Install 2 Packages (+1 Dependent package)

Total download size: 12 M
Installed size: 279 M
Is this ok [y/d/N]: y
Downloading packages:
(1/3): openscap-scanner-1.2.17-11.el7.x86_64.rpm | 63 kB 00:00:02

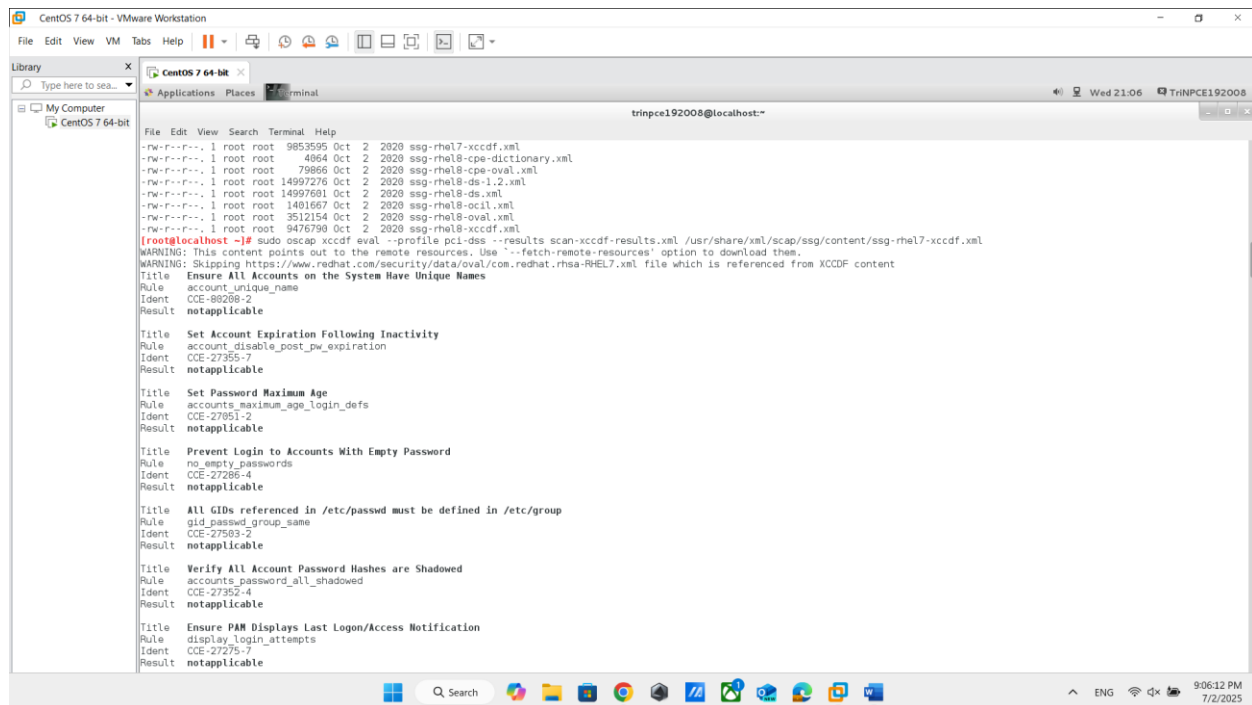
trinpc192008@localhost:~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



2 ảnh trên: cài đặt và xem file profile

Scanning the system:



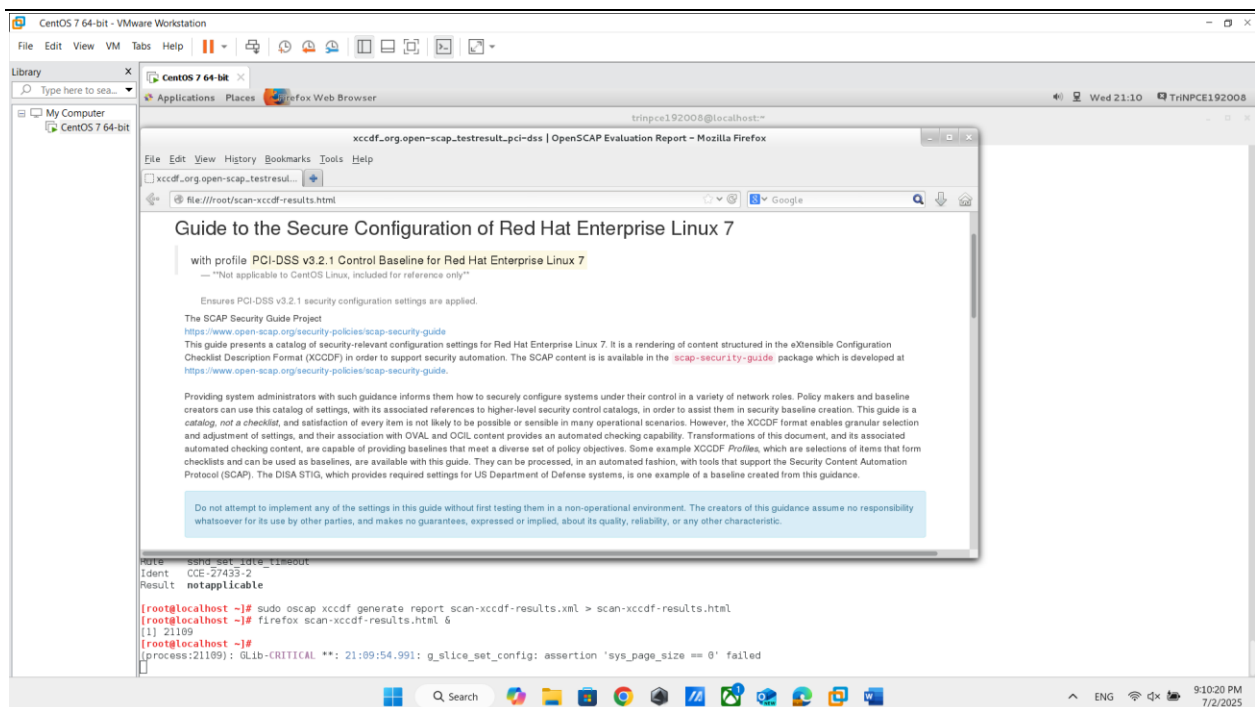
ảnh trên: Quét hệ thống bằng hồ sơ PCI-DSS

oscap xccdf eval: Sử dụng định dạng XCCDF (Extensible Configuration Checklist Description Format) để đánh giá cấu hình hệ thống.

--profile pci-dss: Chọn hồ sơ đánh giá theo tiêu chuẩn PCI DSS (Payment Card Industry Data Security Standard).

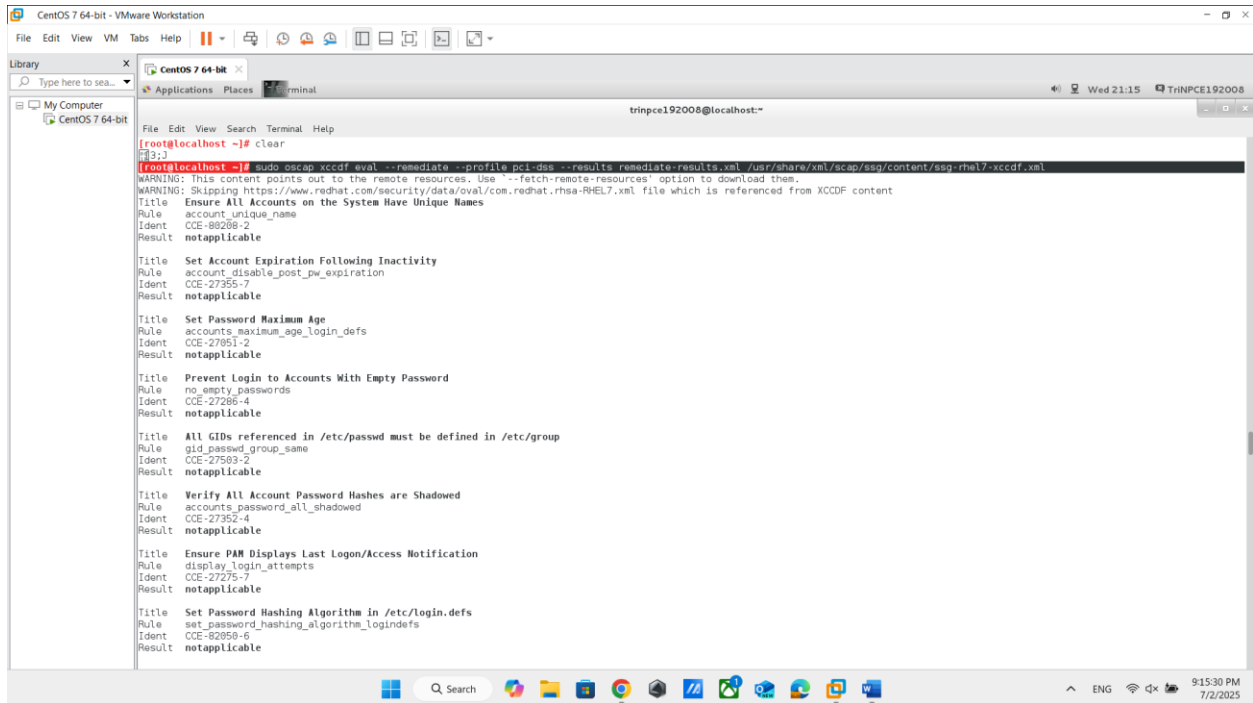
--results scan-xccdf-results.xml: Lưu kết quả quét vào file XML này.

/usr/share/xml/scap/ssg/content/ssg-rhel7-xccdf.xml: Đây là file cấu hình chứa các hồ sơ bảo mật mà hệ thống sẽ dùng.



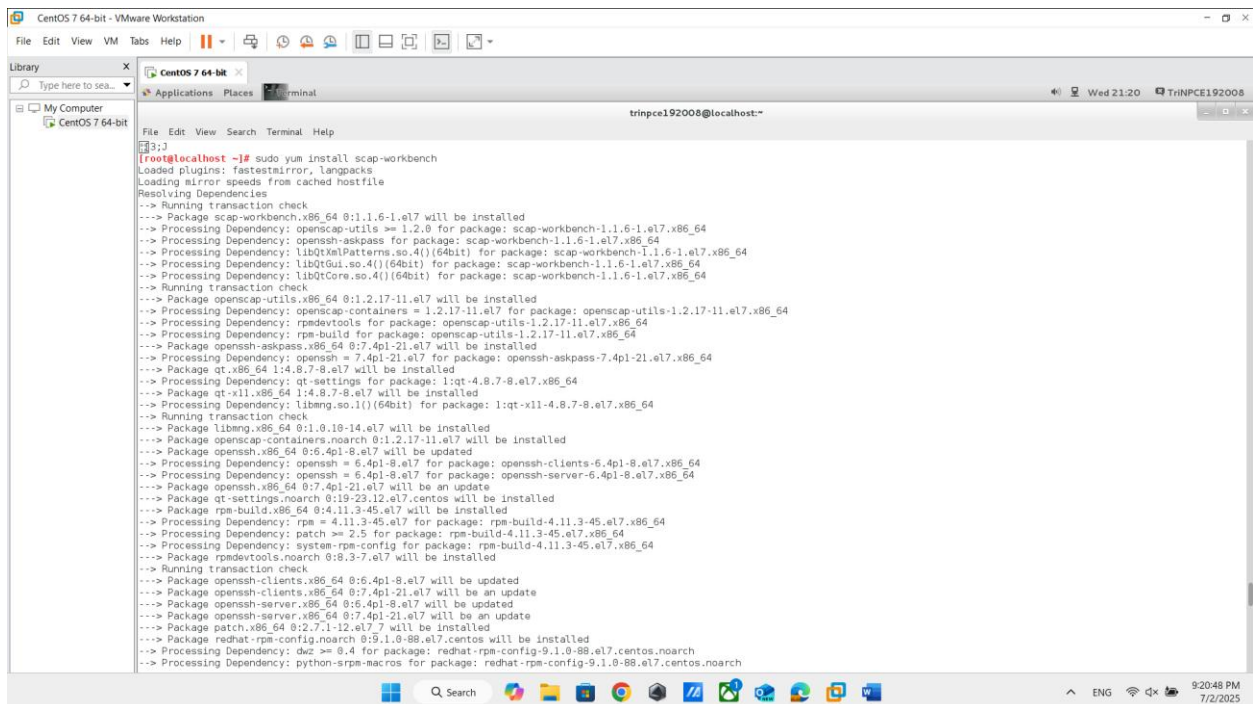
ảnh trên: chuyển file báo cáo xml mà ta đã tạo sang html để dễ đọc

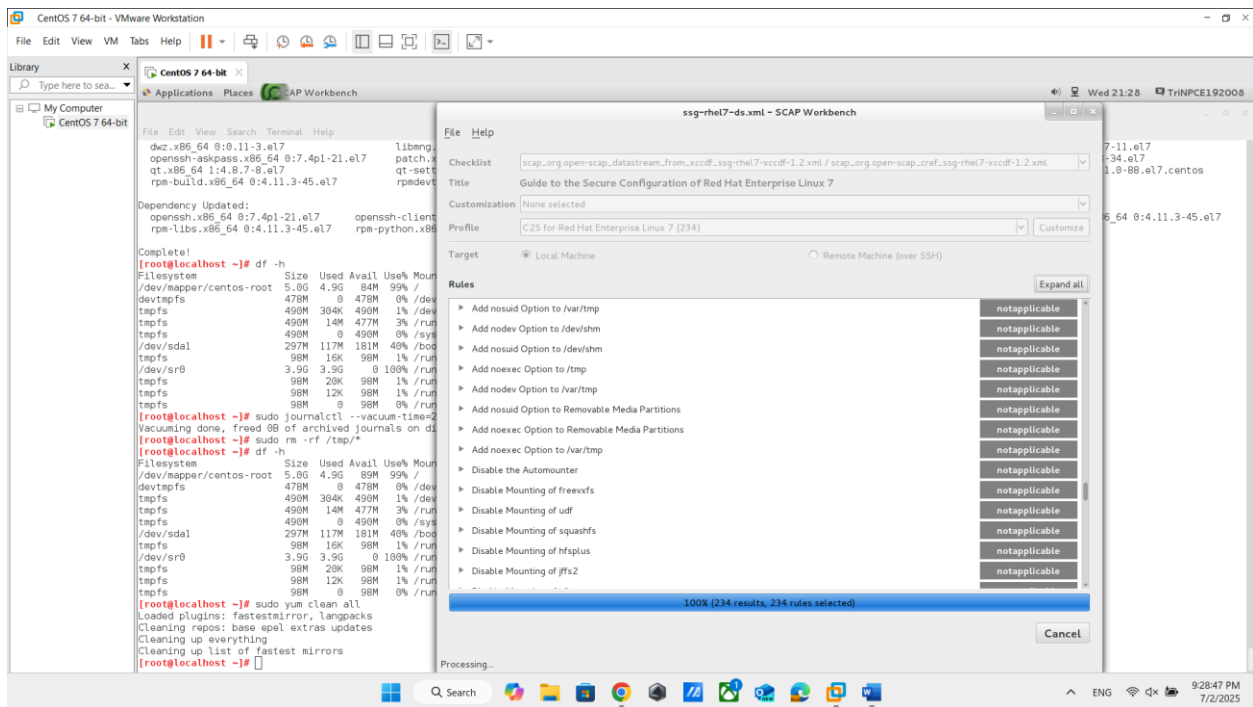
Remediating the system:



ảnh trên: Tự động sửa các vấn đề, lệnh sẽ cố gắng sửa những cấu hình chưa đạt chuẩn

Using SCAP Workbench:





4 ảnh trên: cài đặt và cấu hình sử dụng SCAP Workbench để quét hệ thống